BCA 366

DSE IV Lab — Android Programming

T.Y. B.C.A. (Science) — Semester VI

CBCS 2019 Pattern | Slips 1–20 | All Questions + OR Options

MAEER's MIT Arts, Commerce and Science College, Alandi (D), Pune

Duration: 3 Hours | Max. Marks: 35 + 15 = 50

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Slip 1

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create a Simple Application which shows the Life Cycle of Activity. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.activitylifecycle;

import android.os.Bundle;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Toast.makeText(getApplicationContext(), "onCreate Called", Toast.LENGTH_LONG).show();

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
    @Override
    protected void onStart() {
        super.onStart();
        Toast.makeText(getApplicationContext(), "onStart Called", Toast.LENGTH_LONG).show();
    }
    @Override
```

```
    protected void onRestart() {
        super.onRestart();
        Toast.makeText(getApplicationContext(), "onRestart Called", Toast.LENGTH_LONG).show();
    }
    @Override
    protected void onPause() {
        super.onPause();
        Toast.makeText(getApplicationContext(), "onPause Called", Toast.LENGTH_LONG).show();
    }
     @Override
    protected void onResume() {
        super.onResume();
        Toast.makeText(getApplicationContext(), "onResume Called", Toast.LENGTH_LONG).show();
    }
    @Override
    protected void onStop() {
        super.onStop();
        Toast.makeText(getApplicationContext(), "onStop Called", Toast.LENGTH_LONG).show();
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
       Toast.makeText(getApplicationContext(), "onDestroy Called", Toast.LENGTH_LONG).show();
    }

}
```

Expected Output:

```
On launch: Toast messages shown in sequence:
  onCreate Called → onStart Called → onResume Called
On Back press: onPause Called → onStop Called → onDestroy Called
Logcat (filter: LifeCycle) shows all lifecycle method logs.
```

Q2 — Question

Create an Android application to demonstrate Progress Dialog Box using AsyncTask. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">
    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Download"
        android:textSize="18sp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.asynctaskdemo;

import android.app.ProgressDialog;
import android.os.AsyncTask;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button btn = findViewById(R.id.btnStart);
        btn.setOnClickListener(v -> new DownloadTask().execute());
    }

    class DownloadTask extends AsyncTask<Void, Integer, String> {
        ProgressDialog pd;

        @Override
        protected void onPreExecute() {
            pd = new ProgressDialog(MainActivity.this);
            pd.setTitle("Downloading...");
            pd.setMessage("Please wait...");
            pd.setProgressStyle(ProgressDialog.STYLE_HORIZONTAL);
            pd.setMax(100);
            pd.show();
        }

        @Override
        protected String doInBackground(Void... params) {
            for (int i = 0; i <= 100; i += 10) {
                try { Thread.sleep(500); } catch (InterruptedException e) { e.printStackTrace(); }
                publishProgress(i);
            }
            return "Download Complete!";
        }

        @Override
        protected void onProgressUpdate(Integer... values) {
            pd.setProgress(values[0]);
        }

        @Override
        protected void onPostExecute(String result) {
            pd.dismiss();
            android.widget.Toast.makeText(MainActivity.this, result,
android.widget.Toast.LENGTH_LONG).show();
        }
    }
}
```

Expected Output:

```
Click 'Start Download' → ProgressDialog appears with title "Downloading...".
Progress bar fills from 0% to 100% in increments.
On completion: Dialog dismisses → Toast: "Download Complete!"
```

OR

Q2 — OR (Alternative Question)

Create an Android Application that demonstrate DatePicker and DatePickerDialog. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="221dp"
        android:layout_height="67dp"
        android:text="DatePicker"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.046" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="267dp"
        android:layout_height="57dp"
        android:text="Date"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.132" />

    <Button
        android:id="@+id/button"
        android:layout_width="199dp"
        android:layout_height="70dp"
        android:text="Date"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView3"
        app:layout_constraint Vertical_bias="0.374" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

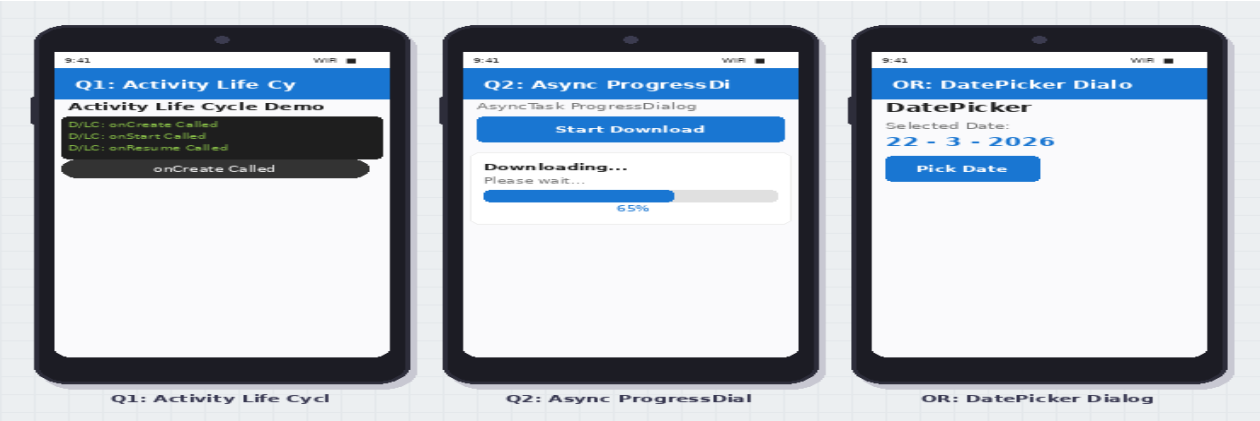
```
package com.example.datepicker;

import android.app.DatePickerDialog;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.DatePicker;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;

//import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
    Button b;
```

```
    TextView tv;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        b=(Button) findViewById(R.id.button);
        tv=(TextView) findViewById(R.id.textView3);

        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                final Calendar c = Calendar.getInstance();

                // on below line we are getting our day, month and year.
                int year = c.get(Calendar.YEAR);
                int month = c.get(Calendar.MONTH);
                int day = c.get(Calendar.DAY_OF_MONTH);

                // on below line we are creating a variable for date picker dialog.
                DatePickerDialog datePickerDialog = new DatePickerDialog(
                        // on below line we are passing context.
                        MainActivity.this,
                        new DatePickerDialog.OnDateSetListener() {
                            @Override
                            public void onDateSet(DatePicker view, int year,
                                                  int monthOfYear, int dayOfMonth) {
                                // on below line we are setting date to our text view.

                                tv.setText(dayOfMonth + "-" + (monthOfYear + 1) + "-" + year);
                            }
                        },
                        // on below line we are passing year,month and day for selected date in our
date picker.
                        year, month, day);
                // at last we are calling show to display our date picker dialog.
                datePickerDialog.show();
            }
        });




        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Click 'Date' button → DatePickerDialog opens.
Select a date → TextView updates: e.g., 22-3-2026.
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)



Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 2

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create a Simple Application which reads a positive number from the user and display its factorial value in another activity. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView2"
        android:layout_width="227dp"
        android:layout_height="59dp"
        android:text="Factorial"
        android:textSize="34sp"
        app:layout_constraintBottom_toTopOf="@+id/editTextText"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="156dp"
        android:ems="10"
        android:inputType="text"
        android:text="Enter a number"
        app:layout_constraintBottom_toTopOf="@+id/button"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.457"
        app:layout_constraintStart_toStartOf="parent" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="316dp"
        android:text="Submit"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_main2.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    tools:context=".MainActivity2">

    <TextView
        android:id="@+id/textView3"
        android:layout_width="222dp"
        android:layout_height="50dp"
        android:text="Factorial is"
        android:textSize="34sp"
        app:layout_constraintBottom_toTopOf="@+id/textView4"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.296" />

    <TextView
        android:id="@+id/textView4"
        android:layout_width="147dp"
        android:layout_height="33dp"
        android:layout_marginBottom="340dp"
        android:text="Answer"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.568"
        app:layout_constraintStart_toStartOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.factorialonnextactivity;

import android.annotation.SuppressLint;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
EditText e1;
Button btn;
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        e1=findViewById(R.id.editTextText);
        btn=findViewById(R.id.button);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String aa = e1.getText().toString();
                Intent i = new Intent(MainActivity.this, MainActivity2.class
```

```
                );
                i.putExtra("no",aa);
                startActivity(i);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

MainActivity2.java

```
package com.example.factorialonnextactivity;

import android.annotation.SuppressLint;
import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity2 extends AppCompatActivity {
    TextView r;
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main2);

        r=findViewById(R.id.textView4);
        Intent i = getIntent();
        int num = Integer.parseInt(i.getStringExtra("no"));
        int p,f = 1;
        for (p=1;p<=num; p++){
            f=f*p;
        }
        r.setText(String.valueOf(f));

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Activity 1: Enter 5 → Click Submit.
Activity 2 displays: "Factorial is  120"
```

Q2 — Question

Create an Android application that plays an audio (song) in the background. Audio will not be stopped even if you switch to another activity. To stop the audio, you need to stop the service. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="224dp"
        android:layout_marginBottom="459dp"
        android:text="Play"
        android:layout_weight="1"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.0"></Button>
    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="224dp"
        android:layout_marginBottom="459dp"
        android:text="Stop"
        android:layout_weight="1"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.0"></Button>

</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.BackgroundMusicfinal"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
```

```
        </activity>
        <service android:name=".MyService"></service>
    </application>

</manifest>
```

MainActivity.java

```
package com.example.backgroundmusicfinal;

import android.annotation.SuppressLint;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    private Button btn1,btn2;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        btn1=findViewById(R.id.btn);
        btn2=findViewById(R.id.btn1);
        btn1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
Intent intent=new Intent(MainActivity.this,MyService.class);
                startService(intent);
            }
        });

        btn2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                stopService(new Intent(MainActivity.this,MyService.class));
            }
        });
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }

}
```

MyService.java

```
package com.example.backgroundmusicfinal;

import android.app.Service;
import android.content.Intent;
import android.media.MediaPlayer;
import android.os.IBinder;
import android.provider.Settings;
```

```
import androidx.annotation.Nullable;

public class MyService extends Service {
    MediaPlayer player;
    @Nullable
    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        //player=MediaPlayer.create(this, Settings.System.DEFAULT_RINGTONE_URI);
        player= MediaPlayer.create(this,R.raw.audio);
        player.start();
        player.setVolume(80.0f,80.0f);
        return START_STICKY;

    }

    @Override
    public void onDestroy() {
        super.onDestroy();
        player.stop();
    }

}
```

Expected Output:

```
Click 'Play' → Music starts playing in background (via Service).
Switch to another app → Music continues playing.
Click 'Stop' → Music stops (Service is stopped).
Note: Place an audio file named 'audio' in res/raw/ folder.
```

OR

Q2 — OR (Alternative Question)

Create an Android Application to display satellite view of current location using Google Map. **[20 Marks]**

Program Code:

MapsActivity.java (activity_maps.xml from Google Maps template)

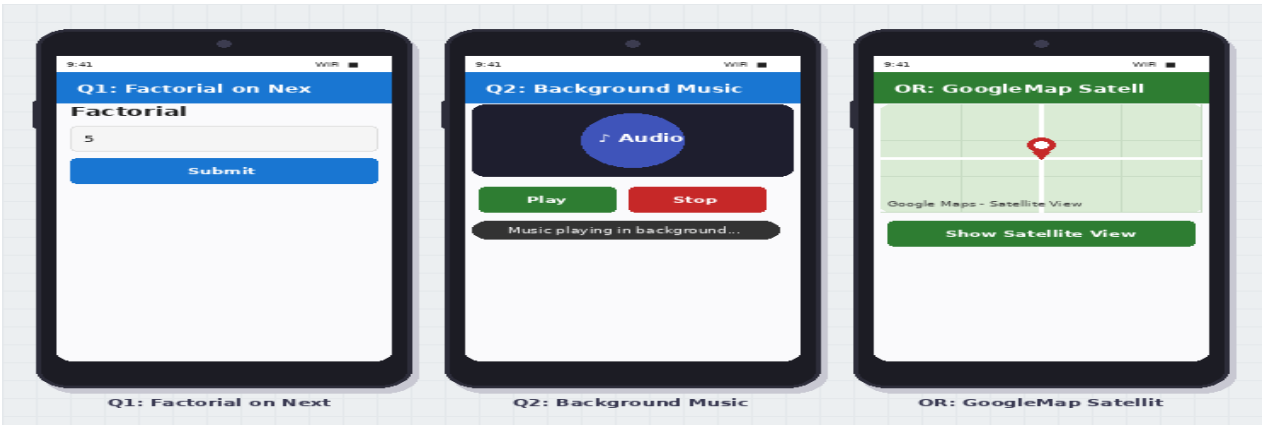
```
package com.example.satellitemap;

import androidx.fragment.app.FragmentActivity;
import android.os.Bundle;
import com.google.android.gms.maps.*;
import com.google.android.gms.maps.model.*;

public class MapsActivity extends FragmentActivity implements OnMapReadyCallback {
    private GoogleMap mMap;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_maps);
        SupportMapFragment mapFragment = (SupportMapFragment)
            getSupportFragmentManager().findFragmentById(R.id.map);
        mapFragment.getMapAsync(this);
    }

    @Override
    public void onMapReady(GoogleMap googleMap) {
```

```
        mMap = googleMap;
        // Set satellite view
        mMap.setMapType(GoogleMap.MAP_TYPE_SATELLITE);
        LatLng pune = new LatLng(18.5204, 73.8567);
        mMap.addMarker(new MarkerOptions().position(pune).title("Pune"));
        mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(pune, 15));
    }
}
```

Expected Output:

```
App opens → Google Map loads in Satellite mode.
Map shows satellite view of the specified location (Pune).
Marker with title "Pune" appears on the map.
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)



Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 3

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create an Android Application that will change color of the College Name on click of Push Button and change the font size, font style of text view using xml. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center">

        <TextView
            android:id="@+id/textView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Font Changes"
            android:textSize="34sp" />
    </TableRow>

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent" >

        <TextView
            android:id="@+id/textView2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Enter Font Size"
            android:textSize="24sp" />

        <EditText
            android:id="@+id/editTextText"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:ems="10"
            android:inputType="text"
            android:textSize="24sp" />
    </TableRow>

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center">

        <Button
            android:id="@+id/button"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Submit"
            android:textSize="34sp" />
```

```
    </TableRow>

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center">

        <TextView
            android:id="@+id/textView3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="MIT ACSC"
            android:textSize="24sp" />
    </TableRow>
</TableLayout>
```

MainActivity.java

```
package com.example.fontchanges;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText e;
    Button b;
    TextView t;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        e=(EditText) findViewById(R.id.editTextText);
        t=(TextView) findViewById(R.id.textView3);
        b=(Button) findViewById(R.id.button);

        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int f=Integer.parseInt(e.getText().toString());
                t.setTextSize(f);
                t.setTextColor(Color.parseColor("#00FFFF"));
            }
        });


        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
```

```
}
```

Expected Output:

```
Enter font size (e.g., 36) → Click Submit.
TextView "MIT ACSC" changes to size 36 and color Cyan (#00FFFF).
```

Q2 — Question

Create an Android Application to find the factorial of a number and Display the Result on Alert Box. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button"
        android:layout_width="287dp"
        android:layout_height="117dp"
        android:text="Factorial Alert"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="274dp"
        android:layout_height="90dp"
        android:ems="10"
        android:inputType="text"
        android:textSize="34sp"
        app:layout_constraintBottom_toTopOf="@+id/button"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.452"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.7" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="208dp"
        android:layout_height="56dp"
        android:text="Factorial"
        android:textSize="34sp"
        app:layout_constraintBottom_toTopOf="@+id/editTextText"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.alertdialogforfactorial;

import android.content.DialogInterface;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Button b;
        EditText et;
        b=findViewById(R.id.button);
        et=findViewById(R.id.editTextText);

        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                AlertDialog.Builder adb = new AlertDialog.Builder(MainActivity.this);
                adb.setTitle("Factorial");

                int n=Integer.parseInt(et.getText().toString());
                int i,f=1;
                for(i=1;i<=n;i++)
                    f=f*i;
                adb.setMessage("Factorial = "+f);
                adb.setCancelable(true);
                adb.setPositiveButton("OK", new DialogInterface.OnClickListener() {
                    @Override
                    public void onClick(DialogInterface dialog, int which) {
                        Toast.makeText(getApplicationContext(),"You pressed
OK",Toast.LENGTH_LONG).show();
                        //finish();
                    }
                });
                adb.setNegativeButton("Cancel", new DialogInterface.OnClickListener() {
                    @Override
                    public void onClick(DialogInterface dialog, int which) {

                        Toast.makeText(getApplicationContext(),"You pressed
Cancel",Toast.LENGTH_LONG).show();
                    }
                });
                adb.setNeutralButton("Retry", new DialogInterface.OnClickListener() {
                    @Override
```

```
                    public void onClick(DialogInterface dialog, int which) {
                        Toast.makeText(getApplicationContext(),"You Presses
Retry",Toast.LENGTH_LONG).show();
                    }
                });

                // Show the dialog
                adb.create();
                adb.show();

            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Enter 5 in EditText → Click 'Factorial Alert'.
AlertDialog appears: Title: "Factorial"  |  Message: "Factorial = 120"
Buttons: Retry / Cancel / OK
```

OR

Q2 — OR (Alternative Question)

Create an Android App — it reads the Students Details (Name, Surname, Class, Gender, Hobbies, Marks) and display all information in another activity in table format on click of Submit button. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">
    <EditText android:id="@+id/etName"    android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Name"/>
    <EditText android:id="@+id/etSurname" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Surname"/>
    <EditText android:id="@+id/etClass"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Class"/>
    <RadioGroup android:id="@+id/rgGender" android:layout_width="match_parent"
android:layout_height="wrap_content" android:orientation="horizontal">
        <RadioButton android:id="@+id/rbMale"   android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Male" android:checked="true"/>
        <RadioButton android:id="@+id/rbFemale" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Female"/>
    </RadioGroup>
    <EditText android:id="@+id/etHobbies" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Hobbies"/>
    <EditText android:id="@+id/etMarks"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Marks" android:inputType="number"/>
    <Button android:id="@+id/btnSubmit"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Submit" android:layout_marginTop="12dp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.studentdetails;

import android.content.Intent;
```

```java
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText etName    = findViewById(R.id.etName);
        EditText etSurname = findViewById(R.id.etSurname);
        EditText etClass   = findViewById(R.id.etClass);
        RadioGroup rgGender = findViewById(R.id.rgGender);
        EditText etHobbies = findViewById(R.id.etHobbies);
        EditText etMarks   = findViewById(R.id.etMarks);

        findViewById(R.id.btnSubmit).setOnClickListener(v -> {
            RadioButton rb = findViewById(rgGender.getCheckedRadioButtonId());
            Intent i = new Intent(this, MainActivity2.class);
            i.putExtra("name",    etName.getText().toString());
            i.putExtra("surname", etSurname.getText().toString());
            i.putExtra("clazz",   etClass.getText().toString());
            i.putExtra("gender",  rb.getText().toString());
            i.putExtra("hobbies", etHobbies.getText().toString());
            i.putExtra("marks",   etMarks.getText().toString());
            startActivity(i);
        });
    }
}
```

activity_main2.xml (table)

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
android:padding="16dp">
    <TableLayout android:id="@+id/tableLayout"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:stretchColumns="1" android:shrinkColumns="1">
    </TableLayout>
</ScrollView>
```

MainActivity2.java

```java
package com.example.studentdetails;

import android.os.Bundle;
import android.view.Gravity;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        android.content.Intent intent = getIntent();
        TableLayout tl = findViewById(R.id.tableLayout);
        String[][] data = {
            {"Field","Value"},
            {"Name",    intent.getStringExtra("name")},
            {"Surname", intent.getStringExtra("surname")},
            {"Class",   intent.getStringExtra("clazz")},
            {"Gender",  intent.getStringExtra("gender")},
            {"Hobbies", intent.getStringExtra("hobbies")},
            {"Marks",   intent.getStringExtra("marks")},
        };
```

```
        boolean header = true;
        for (String[] row : data) {
            TableRow tr = new TableRow(this);
            for (String cell : row) {
                TextView tv = new TextView(this);
                tv.setText(cell);
                tv.setPadding(12,8,12,8);
                tv.setTextSize(15);
                if (header) { tv.setTypeface(null, android.graphics.Typeface.BOLD);
                    tv.setBackgroundColor(0xFF1565C0); tv.setTextColor(0xFFFFFFFF); }
                else { tv.setBackgroundColor(0xFFE3F2FD); tv.setTextColor(0xFF212121); }
                tr.addView(tv);
            }
            tl.addView(tr);
            header = false;
        }
    }
}
```

Expected Output:

```
Activity 1: Fill Name, Surname, Class, Gender, Hobbies, Marks → Submit.
Activity 2: Displays all details in a formatted table with header row (blue).
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 4

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create a Simple Application that performs Arithmetic Operations (Add, Sub, Mult, Div). Use Constraint Layout. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Calculator"
        android:textSize="48sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.05" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="182dp"
        android:layout_height="60dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.48"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.064" />

    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="178dp"
        android:layout_height="63dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText"
        app:layout_constraintVertical_bias="0.048" />

    <Button
        android:id="@+id/button"
        android:layout_width="132dp"
        android:layout_height="87dp"
        android:layout_marginBottom="144dp"
        android:text="Add"
```

```
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.471"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText2"
        app:layout_constraintVertical_bias="0.192" />

    <Button
        android:id="@+id/button2"
        android:layout_width="135dp"
        android:layout_height="86dp"
        android:text="Sub"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.467"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/button"
        app:layout_constraintVertical_bias="0.106" />

    <Button
        android:id="@+id/button3"
        android:layout_width="126dp"
        android:layout_height="74dp"
        android:text="Mult"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/button2"
        app:layout_constraintVertical_bias="0.092" />

    <Button
        android:id="@+id/button4"
        android:layout_width="128dp"
        android:layout_height="72dp"
        android:text="Div"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/button3" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.simplecalculator;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {
    EditText e1,e2;
    Button b1,b2,b3,b4;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        e1=findViewById(R.id.editTextText);
        e2=findViewById(R.id.editTextText2);
        b1=findViewById(R.id.button);
        b2=findViewById(R.id.button2);
        b3=findViewById(R.id.button3);
        b4=findViewById(R.id.button4);


        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int a=Integer.parseInt(e1.getText().toString());
                int b=Integer.parseInt(e2.getText().toString());
                int c=a+b;
                Toast.makeText(getApplicationContext(),"Addition =
"+String.valueOf(c),Toast.LENGTH_LONG).show();
            }
        });

        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int a=Integer.parseInt(e1.getText().toString());
                int b=Integer.parseInt(e2.getText().toString());
                int c=a-b;
                Toast.makeText(getApplicationContext(),"Subtraction =
"+String.valueOf(c),Toast.LENGTH_LONG).show();
            }
        });

        b3.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int a=Integer.parseInt(e1.getText().toString());
                int b=Integer.parseInt(e2.getText().toString());
                int c=a*b;
                Toast.makeText(getApplicationContext(),"Multiplication =
"+String.valueOf(c),Toast.LENGTH_LONG).show();
            }
        });

        b4.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int a=Integer.parseInt(e1.getText().toString());
                int b=Integer.parseInt(e2.getText().toString());
                int c=a/b;
                Toast.makeText(getApplicationContext(),"Division =
"+String.valueOf(c),Toast.LENGTH_LONG).show();
            }
        });
```

```
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Enter two numbers (e.g., 8 and 7).
Click Add → Toast: "Addition = 15"
Click Sub → Toast: "Subtraction = 1"
Click Mult → Toast: "Multiplication = 56"
Click Div → Toast: "Division = 1"
```

Q2 — Question

Create an Android Application that sends the Notification on click of the button and displays the notification message on the second activity. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <Button
        android:id="@+id/btnNotify"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Notification"
        android:textSize="18sp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.notificationdemo;

import android.app.*;
import android.content.Intent;
import android.os.Build;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;

public class MainActivity extends AppCompatActivity {
    static final String CHANNEL_ID = "demo_channel";
    static final int NOTIF_ID = 1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        createNotificationChannel();
        Button btn = findViewById(R.id.btnNotify);
        btn.setOnClickListener(v -> sendNotification());
    }

    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            NotificationChannel channel = new NotificationChannel(
```

```
                CHANNEL_ID, "Demo Channel", NotificationManager.IMPORTANCE_DEFAULT);
            NotificationManager mgr = getSystemService(NotificationManager.class);
            mgr.createNotificationChannel(channel);
        }
    }

    private void sendNotification() {
        Intent intent = new Intent(this, MainActivity.class);
        PendingIntent pi = PendingIntent.getActivity(this, 0, intent,
            PendingIntent.FLAG_IMMUTABLE);
        NotificationCompat.Builder builder = new NotificationCompat.Builder(this, CHANNEL_ID)
            .setSmallIcon(android.R.drawable.ic_dialog_info)
            .setContentTitle("Hello!")
            .setContentText("This is a notification message.")
            .setContentIntent(pi)
            .setAutoCancel(true);
        NotificationManager mgr = (NotificationManager) getSystemService(NOTIFICATION_SERVICE);
        mgr.notify(NOTIF_ID, builder.build());
    }
}
```

Expected Output:

```
Click 'Send Notification' → Notification appears in status bar: Title: "Hello!" | Body: "This is a
notification message."
Tap notification → Opens the app.
```

OR

Q2 — OR (Alternative Question)

Create an android Application for performing the following operation on the table Customer (id, name, address, phno). (use SQLite database) i) Insert New Customer Details. ii) Show All the Customer Details on Toast Message. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <EditText android:id="@+id/etName" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Customer Name"/>
    <EditText android:id="@+id/etAddress" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Address"/>
    <EditText android:id="@+id/etPhone" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Phone Number"
android:inputType="phone"/>
    <Button android:id="@+id/btnInsert" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:text="Insert New Customer"/>
    <Button android:id="@+id/btnShowAll" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:text="Show All Customers"/>
    <TextView android:id="@+id/tvResult" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:padding="8dp" android:textSize="14sp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.sqlitecustomer;

import android.content.ContentValues;
import android.database.Cursor;
import android.database.sqlite.*;
import android.os.Bundle;
import android.widget.*;
```

```java
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    SQLiteDatabase db;
    EditText etName, etAddress, etPhone;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        db = openOrCreateDatabase("CustomerDB", MODE_PRIVATE, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS Customer " +
            "(id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT, address TEXT, phno TEXT)");

        etName    = findViewById(R.id.etName);
        etAddress = findViewById(R.id.etAddress);
        etPhone   = findViewById(R.id.etPhone);
        tvResult  = findViewById(R.id.tvResult);

        findViewById(R.id.btnInsert).setOnClickListener(v -> {
            ContentValues cv = new ContentValues();
            cv.put("name",    etName.getText().toString());
            cv.put("address", etAddress.getText().toString());
            cv.put("phno",    etPhone.getText().toString());
            db.insert("Customer", null, cv);
            Toast.makeText(this, "Customer inserted!", Toast.LENGTH_SHORT).show();
            etName.setText(""); etAddress.setText(""); etPhone.setText("");
        });

        findViewById(R.id.btnShowAll).setOnClickListener(v -> {
            Cursor c = db.rawQuery("SELECT * FROM Customer", null);
            StringBuilder sb = new StringBuilder();
            while (c.moveToNext())
                sb.append(c.getInt(0)).append(". ").append(c.getString(1))
                  .append(" | ").append(c.getString(2))
                  .append(" | ").append(c.getString(3)).append("\n");
            c.close();
            tvResult.setText(sb.length() > 0 ? sb : "No records.");
        });
    }
}
```

Expected Output:

```
Fill name, address, phone → Click "Insert New Customer" → Toast: "Customer inserted!"
Click "Show All Customers" → TextView shows: "1. Alice | Pune | 9876543210"
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 5

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create a Simple Application Which Sends "Hello!" message from one activity to another with help of Button (Use Intent). **[10 Marks]**

Program Code:

activity_main.xml (Activity 1)

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Hello!"
        android:textSize="20sp"/>
</LinearLayout>
```

MainActivity.java (Activity 1)

```java
package com.example.intenthello;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button btn = findViewById(R.id.btnSend);
        btn.setOnClickListener(v -> {
            Intent i = new Intent(this, MainActivity2.class);
            i.putExtra("message", "Hello!");
            startActivity(i);
        });
    }
}
```

activity_main2.xml (Activity 2)

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <TextView
        android:id="@+id/tvMsg"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="48sp"
        android:textStyle="bold"
        android:textColor="#1565C0"/>
</LinearLayout>
```

MainActivity2.java (Activity 2)

```java
package com.example.intenthello;

import android.os.Bundle;
```

```
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        TextView tv = findViewById(R.id.tvMsg);
        String msg = getIntent().getStringExtra("message");
        tv.setText(msg);
    }
}
```

Expected Output:

```
Activity 1: Click "Send Hello!" button.
Activity 2 opens and displays "Hello!" in large text.
```

Q2 — Question

Create an Android Application that Demonstrates ListView and onClick of List item Display the Toast. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ListView
        android:id="@+id/listview"
        android:layout_width="409dp"
        android:layout_height="729dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.listview;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {
    ListView l;
    String tutorials[] = { "Algorithms",
            "Data Structures",
            "Languages",
            "Interview Corner",
            "GATE",
            "ISRO CS",
            "UGC NET CS",
            "CS Subjects",
            "Web Technologies"};

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        l = findViewById(R.id.listview); //support_simple_spinner_dropdown_item
        ArrayAdapter<String> arr= new ArrayAdapter<String>(this,
                androidx.appcompat.R.layout.support_simple_spinner_dropdown_item,tutorials);
        l.setAdapter(arr);

        l.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position,
                                    long id) {
                // Display a toast message with the clicked item text
                Toast.makeText(MainActivity.this, " You Clicked: " +
                        tutorials[position], Toast.LENGTH_SHORT).show();
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
ListView shows: Algorithms, Data Structures, Languages, Interview Corner, GATE, ISRO CS...
Click any item → Toast: "You Clicked: Algorithms" (or whichever is clicked)
```

OR

Q2 — OR (Alternative Question)

Create an Android application to perform following operations on table Student (Sid, Sname, phno). Use autoincrement for Sid. a) Add Student and display its information. b) Delete Student **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <EditText android:id="@+id/etSname" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Student Name"/>
```

```
    <EditText android:id="@+id/etPhone" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Phone Number"
android:inputType="phone"/>
    <Button android:id="@+id/btnAdd" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:text="Add Student"/>
    <Button android:id="@+id/btnDelete" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:text="Delete Student (by Name)"/>
    <Button android:id="@+id/btnShow" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:text="Show All Students"/>
    <TextView android:id="@+id/tvResult" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:padding="8dp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.sqlitestudent;

import android.content.ContentValues;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    SQLiteDatabase db;
    EditText etSname, etPhone;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        db = openOrCreateDatabase("StudentDB", MODE_PRIVATE, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS Student " +
            "(Sid INTEGER PRIMARY KEY AUTOINCREMENT, Sname TEXT, phno TEXT)");

        etSname  = findViewById(R.id.etSname);
        etPhone  = findViewById(R.id.etPhone);
        tvResult = findViewById(R.id.tvResult);

        findViewById(R.id.btnAdd).setOnClickListener(v -> {
            ContentValues cv = new ContentValues();
            cv.put("Sname", etSname.getText().toString());
            cv.put("phno",  etPhone.getText().toString());
            db.insert("Student", null, cv);
            Toast.makeText(this, "Student added!", Toast.LENGTH_SHORT).show();
            etSname.setText(""); etPhone.setText("");
        });

        findViewById(R.id.btnDelete).setOnClickListener(v -> {
            int rows = db.delete("Student", "Sname=?",
                new String[]{etSname.getText().toString()});
            Toast.makeText(this, rows > 0 ? "Deleted!" : "Not found", Toast.LENGTH_SHORT).show();
        });

        findViewById(R.id.btnShow).setOnClickListener(v -> {
            Cursor c = db.rawQuery("SELECT * FROM Student", null);
            StringBuilder sb = new StringBuilder();
            while (c.moveToNext())
                sb.append(c.getInt(0)).append(". ").append(c.getString(1))
                  .append(" | ").append(c.getString(2)).append("\n");
            c.close();
```

```
            tvResult.setText(sb.length() > 0 ? sb : "No records.");
        });
    }
}
```

Expected Output:

```
Enter name + phone → "Add Student" → Toast: "Student added!"
Enter name → "Delete Student" → Toast: "Deleted!" or "Not found"
"Show All Students" → Displays list of all students.
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 6

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create an Android Application that Demonstrate Radio Button. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical"
    android:layout_marginLeft="50dp"
    android:layout_marginRight="50dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Radio Button and checkbox Demo"
        android:layout_marginTop="50dp"></TextView>
    <RadioGroup
        android:layout_width="match_parent"
        android:layout_height="wrap_content"

        android:orientation="vertical"

        android:layout_marginTop="50dp">

        <RadioButton
            android:id="@+id/rbtFy"
            android:layout_width="166dp"
            android:layout_height="wrap_content"
            android:text="FYBBA CA"></RadioButton>

        <RadioButton
            android:id="@+id/rbtSy"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="SYBBA CA"></RadioButton>

        <RadioButton
            android:id="@+id/rbtTy"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="TYBBA CA"></RadioButton>
    </RadioGroup>

    <CheckBox
        android:id="@+id/chk1"
        android:layout_width="132dp"
        android:layout_height="wrap_content"
        android:text="CPP"></CheckBox>

    <CheckBox
        android:id="@+id/chk2"
        android:layout_width="129dp"
        android:layout_height="wrap_content"
        android:text="DS"></CheckBox>
```

```
    <CheckBox
        android:id="@+id/chk3"
        android:layout_width="174dp"
        android:layout_height="wrap_content"
        android:text="JAVA"></CheckBox>

    <TableLayout
        android:layout_width="match_parent"
        android:layout_height="71dp">

        <TableRow
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">

            <Button
                android:id="@+id/btDisplay"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:text="Display" />

            <Button
                android:id="@+id/btReset"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:text="Reset" />
        </TableRow>
    </TableLayout>

</LinearLayout>
```

MainActivity.java

```
package com.example.radiobuttoncheckbox;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.RadioButton;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        RadioButton bt1,bt2,bt3;
        CheckBox ck1,ck2,ck3;
        Button display,reset;

        bt1=findViewById(R.id.rbtFy);
```

```
        bt2=findViewById(R.id.rbtSy);
        bt3=findViewById(R.id.rbtTy);
        display=findViewById(R.id.btDisplay);
        ck1=findViewById(R.id.chk1);
        ck2=findViewById(R.id.chk2);
        ck3=findViewById(R.id.chk3);
        reset=findViewById(R.id.btReset);

        display.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                String msg1="",msg2="";
                if(bt1.isChecked())
                {
                    msg1="FY BBA CA";
                }
                if(bt2.isChecked())
                {
                    msg1="SY BBA CA";
                }
                if(bt3.isChecked())
                {
                    msg1="TY BBA CA";
                }

                if(ck1.isChecked())
                {
                    msg2=" CPP";
                }
                if(ck2.isChecked())
                {
                    msg2=msg2+" DS";
                }
                if(ck3.isChecked())
                {
                    msg2=msg2+" JAVA";
                }
                Toast.makeText(getApplicationContext(),msg1+"\n"+msg2,Toast.LENGTH_LONG).show();
            }
        });

        reset.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v)
            {
                ck1.setChecked(false);
                ck2.setChecked(false);
                ck3.setChecked(false);
                bt1.setChecked(false);
                bt2.setChecked(false);
                bt3.setChecked(false);
            }
        });
    }
}
```

Expected Output:

```
Select "Male" radio button → Click Submit.
Toast: "Selected: Male"
(Change to Female → Toast: "Selected: Female")
```

Q2 — Question

Create an Android application to demonstrate phone call using Implicit Intent. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="318dp"
        android:layout_height="69dp"
        android:text="Phone Calling"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.827"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.078" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="282dp"
        android:layout_height="66dp"
        android:text="Enter Mobile No"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.637"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.089" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="294dp"
        android:layout_height="60dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.572"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView2"
        app:layout_constraintVertical_bias="0.052" />

    <Button
        android:id="@+id/button"
        android:layout_width="207dp"
        android:layout_height="73dp"
        android:text="Dial"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.544"
```

```
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText"
        app:layout_constraintVertical_bias="0.133" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-feature
        android:name="android.hardware.telephony"
        android:required="false" />
    <uses-permission android:name="android.permission.CALL_PHONE" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.PhoneCall"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

```
package com.example.phonecall;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        EditText mno;
        Button call;
```

```
        mno = findViewById(R.id.editTextText);
        call = findViewById(R.id.button);
        call.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String phone_number = mno.getText().toString();
                 // Getting instance of Intent with action as ACTION_CALL
                Intent phone_intent = new Intent(Intent.ACTION_CALL);
                // Set data of Intent through Uri by parsing phone number
                phone_intent.setData(Uri.parse("tel:" + phone_number));
                // start Intent
                startActivity(phone_intent);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Enter mobile number (e.g., 9876543210) → Click 'Dial'.
Phone dialer opens with the number pre-filled.
Permission: android.permission.CALL_PHONE required.
```

OR

Q2 — OR (Alternative Question)

Write an android code to turn ON/OFF the Wi-Fi. **[20 Marks]**

Program Code:

AndroidManifest.xml

```
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE"/>
<uses-permission android:name="android.permission.CHANGE_WIFI_STATE"/>
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.WAKE_LOCK" />
<uses-permission android:name="android.permission.UPDATE_DEVICE_STATS"
    tools:ignore="ProtectedPermissions" />
```

MainActivity.java

```
package com.example.wifionoff;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Context;
import android.net.wifi.WifiManager;
import android.os.Bundle;
import android.widget.CompoundButton;
import android.widget.TextView;
import android.widget.ToggleButton;

public class MainActivity extends AppCompatActivity {

    ToggleButton toggleButton;
    TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        // Getting toggle button and textView from activity_main
        toggleButton = (ToggleButton) findViewById(R.id.toggleButton);
        textView = (TextView) findViewById(R.id.textView);

        // Put listener on toggle button
        toggleButton.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(CompoundButton compoundButton, boolean checked) {
                if (checked) {
                    textView.setText("WiFi is ON");
                    WifiManager wifi = (WifiManager) getSystemService(Context.WIFI_SERVICE);
                    wifi.setWifiEnabled(true);
                } else {
                    textView.setText("WiFi is OFF");
                    WifiManager wifi = (WifiManager) getSystemService(Context.WIFI_SERVICE);
                    wifi.setWifiEnabled(false);
                }
            }
        });
        // For initial setting
        if (toggleButton.isChecked()) {
            textView.setText("WiFi is ON");
            WifiManager wifi = (WifiManager) getSystemService(Context.WIFI_SERVICE);
            wifi.setWifiEnabled(true);
        } else {
            textView.setText("WiFi is OFF");
            WifiManager wifi = (WifiManager) getSystemService(Context.WIFI_SERVICE);
            wifi.setWifiEnabled(false);
        }


    }
}
```

Expected Output:

```
Toggle ON → TextView shows "WiFi is ON" and device WiFi enables.
Toggle OFF → TextView shows "WiFi is OFF" and device WiFi disables.
Permissions required: ACCESS_WIFI_STATE, CHANGE_WIFI_STATE
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 7

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create an Android App with Login Screen. On successful login, give message and go to next Activity (Without Using Database — use Table Layout). **[10 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="183dp"
        android:layout_height="65dp"
        android:text="Login"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.674"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.058" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="135dp"
        android:layout_height="26dp"
        android:layout_marginBottom="336dp"
        android:text="Username"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.445"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.081" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.417"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView2"
        app:layout_constraintVertical_bias="0.098" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="196dp"
        android:layout_height="46dp"
```

```
        android:text="Password"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText"
        app:layout_constraintVertical_bias="0.095" />

    <Button
        android:id="@+id/button"
        android:layout_width="122dp"
        android:layout_height="76dp"
        android:layout_marginBottom="76dp"
        android:text="Login"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.47"
        app:layout_constraintStart_toStartOf="parent" />

    <EditText
        android:id="@+id/editTextTextPassword"
        android:layout_width="211dp"
        android:layout_height="54dp"
        android:layout_marginBottom="140dp"
        android:ems="10"
        android:inputType="textPassword"
        app:layout_constraintBottom_toTopOf="@+id/button"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.381"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView3"
        app:layout_constraintVertical_bias="1.0" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">

    <TextView
        android:id="@+id/textView5"
        android:layout_width="248dp"
        android:layout_height="81dp"
        android:text="Login Message"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.711"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.12" />

    <TextView
        android:id="@+id/textView6"
        android:layout_width="321dp"
```

```
        android:layout_height="83dp"
        android:layout_marginBottom="300dp"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.602"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView5"
        app:layout_constraintVertical_bias="0.394" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.loginonnextactivity;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText a,b;
    Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        a = (EditText) findViewById(R.id.editTextText);
        b = (EditText) findViewById(R.id.editTextTextPassword);
        btn = (Button) findViewById(R.id.button);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String aaa = a.getText().toString();
                String bbb = b.getText().toString();

                Intent i = new Intent(MainActivity.this, MainActivity2.class);
                i.putExtra("uname", aaa);
                i.putExtra("password", bbb);
                startActivity(i);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

MainActivity2.java

```
package com.example.loginonnextactivity;

import android.content.Intent;
import android.os.Bundle;
```

```java
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity2 extends AppCompatActivity {
    TextView v;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main2);
        v=findViewById(R.id.textView6);
        Intent i=getIntent();
        String a=i.getStringExtra("uname");
        String b= i.getStringExtra("password");
        if(a.equals("Admin")&&b.equals("123"))
         v.setText(String.valueOf("Welcome Admin"));
        else
            v.setText(String.valueOf("Invalid Login"));

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Enter Username: Admin, Password: 123 → Click Login.
Activity 2 shows: "Welcome Admin"
Enter wrong credentials → Activity 2: "Invalid Login"
```

Q2 — Question

Create android application to demonstrate how to use a service to download a file from the Internet on click of Download Button. Once done, the service notifies the activity via a broadcast receiver that the download is complete. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:gravity="center" android:orientation="vertical" android:padding="16dp">
    <Button android:id="@+id/btnDownload"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Start Download" android:textSize="18sp"/>
    <TextView android:id="@+id/tvStatus"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Status: Idle" android:textSize="16sp" android:layout_marginTop="20dp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.downloadservice;

import android.content.*;
import android.os.Bundle;
import android.widget.*;
```

```java
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    TextView tvStatus;
    BroadcastReceiver receiver = new BroadcastReceiver() {
        @Override
        public void onReceive(Context context, Intent intent) {
            tvStatus.setText("Status: " + intent.getStringExtra("status"));
        }
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        tvStatus = findViewById(R.id.tvStatus);
        Button btn = findViewById(R.id.btnDownload);
        btn.setOnClickListener(v -> {
            tvStatus.setText("Status: Downloading...");
            startService(new Intent(this, DownloadService.class));
        });
    }

    @Override protected void onResume() {
        super.onResume();
        registerReceiver(receiver, new IntentFilter("DOWNLOAD_COMPLETE"));
    }
    @Override protected void onPause() {
        super.onPause();
        unregisterReceiver(receiver);
    }
}
```

📄 **DownloadService.java**

```java
package com.example.downloadservice;

import android.app.Service;
import android.content.*;
import android.os.*;
import android.util.Log;

public class DownloadService extends Service {
    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        new Thread(() -> {
            try {
                // Simulate download
                Thread.sleep(3000);
            } catch (InterruptedException e) { e.printStackTrace(); }
            // Notify Activity via Broadcast
            Intent broadcast = new Intent("DOWNLOAD_COMPLETE");
            broadcast.putExtra("status", "Download Complete!");
            sendBroadcast(broadcast);
            stopSelf();
        }).start();
        return START_STICKY;
    }
    @Override public IBinder onBind(Intent intent) { return null; }
}
```

Expected Output:

```
Click "Start Download" → Status: "Downloading..."
After 3 seconds → BroadcastReceiver notifies Activity.
Status updates: "Download Complete!"
```

OR

Q2 — OR (Alternative Question)

Create application to send email with attachment. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="129dp"
        android:layout_height="49dp"
        android:text="Email"
        android:textColor="#D11B1B"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.507"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.049" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="88dp"
        android:layout_height="35dp"
        android:text="To"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.47"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.13" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="285dp"
        android:layout_height="52dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.532"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.204" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="126dp"
        android:layout_height="40dp"
        android:text="Subject"
        android:textSize="24sp"
```

```
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.29" />

    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="283dp"
        android:layout_height="51dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.532"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.379" />

    <TextView
        android:id="@+id/textView4"
        android:layout_width="157dp"
        android:layout_height="40dp"
        android:text="Body"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />

    <EditText
        android:id="@+id/editTextText3"
        android:layout_width="278dp"
        android:layout_height="54dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.532"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.618" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Email"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraint End_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.505"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.813" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
```

```
    <uses-permission android:name="android.permission.INTERNET" />
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Email"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

```
package com.example.email;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Button send;
        EditText to, subject, body;
        to=(EditText) findViewById(R.id.editTextText);
        subject=(EditText) findViewById(R.id.editTextText2);
        body=(EditText) findViewById(R.id.editTextText3);
        send=(Button) findViewById(R.id.button);
        send.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String emailto = to.getText().toString();
                String emailsubject = subject.getText().toString();
                String emailbody = body.getText().toString();
                // define Intent object with action attribute as ACTION_SEND
                Intent intent = new Intent(Intent.ACTION_SEND);
                // add three fields to intent using putExtra function
```

```
                intent.putExtra(Intent.EXTRA_EMAIL, new String[]{emailto});
                intent.putExtra(Intent.EXTRA_SUBJECT, emailsubject);
                intent.putExtra(Intent.EXTRA_TEXT, emailbody);
                // set type of intent
                intent.setType("message/rfc822");
  // startActivity with intent with chooser as Email client using createChooser function
                startActivity(Intent.createChooser(intent, "Choose an Email client :"));
            }
        });


        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Fill To, Subject, Body → Click "Send Email".
Email chooser dialog opens → Select Gmail/Outlook.
Email client opens with all fields pre-filled.
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 8

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Write an Android application to accept two numbers from the user, and display them, but reject input if both numbers are greater than 10 and asks for two new numbers. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="20dp" android:gravity="center">
    <EditText android:id="@+id/etNum1" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Enter first number"
        android:inputType="number"/>
    <EditText android:id="@+id/etNum2" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Enter second number"
        android:inputType="number" android:layout_marginTop="10dp"/>
    <Button android:id="@+id/btnSubmit" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Submit"
        android:layout_marginTop="16dp"/>
    <TextView android:id="@+id/tvResult" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:textSize="16sp"
        android:layout_marginTop="16dp" android:textColor="#E65100"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.twonumbers;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText etNum1, etNum2;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        etNum1   = findViewById(R.id.etNum1);
        etNum2   = findViewById(R.id.etNum2);
        tvResult = findViewById(R.id.tvResult);

        findViewById(R.id.btnSubmit).setOnClickListener(v -> {
            int n1 = Integer.parseInt(etNum1.getText().toString());
            int n2 = Integer.parseInt(etNum2.getText().toString());
            if (n1 > 10 && n2 > 10) {
                tvResult.setText("Both numbers > 10! Please enter new numbers.");
                etNum1.setText(""); etNum2.setText("");
            } else {
                tvResult.setText("Accepted: " + n1 + " and " + n2);
            }
        });
    }
}
```

Expected Output:

```
Enter 12 and 15 → Click Submit → Error: "Both numbers > 10! Please enter new numbers." (fields cleared)
Enter 5 and 12 → Click Submit → "Accepted: 5 and 12"
```

Q2 — Question

Write a program to find the specific location of an Android device and display details of the place like Address line, city with Geocoding. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp" android:gravity="center">
    <Button android:id="@+id/btnGetLocation" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Get Current Location"
android:textSize="16sp"/>
    <TextView android:id="@+id/tvLocation" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:textSize="14sp" android:padding="16dp"
        android:layout_marginTop="16dp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.gpslocation;

import android.Manifest;
import android.content.pm.PackageManager;
import android.location.*;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import java.io.IOException;
import java.util.*;

public class MainActivity extends AppCompatActivity {
    TextView tvLocation;
    LocationManager lm;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        tvLocation = findViewById(R.id.tvLocation);
        lm = (LocationManager) getSystemService(LOCATION_SERVICE);
        ActivityCompat.requestPermissions(this,
            new String[]{Manifest.permission.ACCESS_FINE_LOCATION}, 1);

        findViewById(R.id.btnGetLocation).setOnClickListener(v -> getLocation());
    }

    private void getLocation() {
        if (ActivityCompat.checkSelfPermission(this,
                Manifest.permission.ACCESS_FINE_LOCATION) != PackageManager.PERMISSION_GRANTED) {
            tvLocation.setText("Location permission not granted.");
            return;
        }
        Location loc = lm.getLastKnownLocation(LocationManager.GPS_PROVIDER);
        if (loc == null) loc = lm.getLastKnownLocation(LocationManager.NETWORK_PROVIDER);
        if (loc != null) {
            double lat = loc.getLatitude(), lng = loc.getLongitude();
            Geocoder gc = new Geocoder(this, Locale.getDefault());
            try {
                List<Address> addresses = gc.getFromLocation(lat, lng, 1);
                if (!addresses.isEmpty()) {
```

```
                    Address addr = addresses.get(0);
                    tvLocation.setText("Address: " + addr.getAddressLine(0) +
                        "\nCity: " + addr.getLocality() +
                        "\nLat: " + lat + "  Lng: " + lng);
                }
            } catch (IOException e) {
                tvLocation.setText("Lat: " + lat + "\nLng: " + lng);
            }
        } else {
            tvLocation.setText("Location unavailable. Enable GPS.");
        }
    }
}
```

Expected Output:

```
Click "Get Current Location" → Fetches GPS coordinates.
Displays: Address: Alandi, Pune, Maharashtra
City: Pune  |  Lat: 18.6783  Lng: 73.8950
Requires: ACCESS_FINE_LOCATION permission.
```

OR

Q2 — OR (Alternative Question)

Create table Company (id, name, address, phno). Create Application for Performing the following operation on the table. a) Insert New Company details. b) Show All Company details **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <EditText android:id="@+id/etName" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Company Name"/>
    <EditText android:id="@+id/etAddress" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Address"/>
    <EditText android:id="@+id/etPhone" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Phone Number"
android:inputType="phone"/>
    <Button android:id="@+id/btnInsert" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:text="Insert New Customer"/>
    <Button android:id="@+id/btnShowAll" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:text="Show All Customers"/>
    <TextView android:id="@+id/tvResult" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:padding="8dp" android:textSize="14sp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.sqlitecompany;

import android.content.ContentValues;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    SQLiteDatabase db;
    EditText etName, etAddress;
    TextView tvResult;
```

```java
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        db = openOrCreateDatabase("CompanyDB", MODE_PRIVATE, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS Company " +
            "(id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT, address TEXT, phno TEXT)");
        etName    = findViewById(R.id.etName);
        etAddress = findViewById(R.id.etAddress);
        tvResult  = findViewById(R.id.tvResult);

        findViewById(R.id.btnInsert).setOnClickListener(v -> {
            ContentValues cv = new ContentValues();
            cv.put("name",    etName.getText().toString());
            cv.put("address", etAddress.getText().toString());
            db.insert("Company", null, cv);
            Toast.makeText(this, "Company inserted!", Toast.LENGTH_SHORT).show();
        });

        findViewById(R.id.btnShowAll).setOnClickListener(v -> {
            Cursor c = db.rawQuery("SELECT * FROM Company", null);
            StringBuilder sb = new StringBuilder();
            while (c.moveToNext())
                sb.append(c.getInt(0)).append(". ").append(c.getString(1))
                  .append(" | ").append(c.getString(2)).append("\n");
            c.close();
            tvResult.setText(sb.length() > 0 ? sb : "No records.");
        });
    }
}
```

Expected Output:

```
Enter company details → "Insert Company" → Toast: "Company inserted!"
"Show All Companies" → TextView shows all records.
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 9

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create an Android Application that Demonstrate Switch and Toggle Button. **[10 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical"
    >

    <Switch
        android:id="@+id/simpleSwitch"
        android:layout_width="138dp"
        android:layout_height="12dp"
        android:checked="true"
        android:textOff="off"
        android:textOn="onn"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.442"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.086" />

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

        android:checked="true"
        android:text="ToggleButton"
        android:textOff="Off"
        android:textOn="On"

        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/simpleSwitch"
        app:layout_constraintVertical_bias="0.228" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="340dp"
        android:layout_height="52dp"
        android:textSize="20sp"
        app:layout_constraintBottom_toTopOf="@+id/toggleButton"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/simpleSwitch" />
```

```
    <TextView
        android:layout_width="330dp"
        android:layout_height="49dp"
        android:text=""
        android:id="@+id/tb2"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.496"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/toggleButton"
        app:layout_constraintVertical_bias="0.121"></TextView>


</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.togglebuttonswitchdemo;

import android.os.Bundle;
import android.view.View;
import android.widget.CompoundButton;
import android.widget.Switch;
import android.widget.TextView;
import android.widget.ToggleButton;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    Switch simpleSwitch;
    ToggleButton toggleButton;
    TextView switchtv,togglebuttontv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        simpleSwitch = (Switch) findViewById(R.id.simpleSwitch);
        toggleButton=(ToggleButton)findViewById(R.id.toggleButton);
        switchtv=findViewById(R.id.textView);
        togglebuttontv=findViewById(R.id.tb2);

        toggleButton.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(CompoundButton compoundButton, boolean checked) {
                if (checked) {
                    togglebuttontv.setText("Toggle button is ON");

                } else {
                    togglebuttontv.setText("Toggle button is OFF");
                }
            }
        });

        simpleSwitch.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(CompoundButton compoundButton, boolean checked) {
```

```
                if (checked) {
                    switchtv.setText("Switch button is ON");

                } else {
                    switchtv.setText("Switch button is OFF");

                }
            }
        });



        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Switch toggled ON → TextView: "Switch button is ON"
Switch toggled OFF → TextView: "Switch button is OFF"
ToggleButton pressed ON → TextView: "Toggle button is ON"
ToggleButton pressed OFF → TextView: "Toggle button is OFF"
```

Q2 — Question

Create a fragment that has its own UI and enable your activities to communicate with fragments. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">
    <Button android:id="@+id/btnLoadFragment"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Load Fragment"/>
    <FrameLayout android:id="@+id/fragmentContainer"
        android:layout_width="match_parent" android:layout_height="match_parent"
        android:layout_marginTop="16dp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.fragmentdemo;

import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button btn = findViewById(R.id.btnLoadFragment);
        btn.setOnClickListener(v -> {
            MyFragment fragment = new MyFragment();
            FragmentTransaction ft = getSupportFragmentManager().beginTransaction();
            ft.replace(R.id.fragmentContainer, fragment);
```

```
            ft.commit();
        });
    }
}
```

fragment_my.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp" android:background="#E3F2FD">
    <TextView android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Hello from Fragment!" android:textSize="20sp" android:textColor="#1565C0"
        android:textStyle="bold"/>
    <Button android:id="@+id/btnFragAction"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Click Me" android:layout_marginTop="12dp"/>
</LinearLayout>
```

MyFragment.java

```
package com.example.fragmentdemo;

import android.os.Bundle;
import android.view.*;
import android.widget.*;
import androidx.fragment.app.Fragment;

public class MyFragment extends Fragment {
    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
                             Bundle savedInstanceState) {
        View v = inflater.inflate(R.layout.fragment_my, container, false);
        Button btn = v.findViewById(R.id.btnFragAction);
        btn.setOnClickListener(b ->
            Toast.makeText(getContext(), "Fragment Button Clicked!", Toast.LENGTH_SHORT).show());
        return v;
    }
}
```

Expected Output:

```
Click "Load Fragment" → Fragment UI appears in FrameLayout.
Fragment shows: "Hello from Fragment!" text and a button.
Click Fragment button → Toast: "Fragment Button Clicked!"
```

OR

Q2 — OR (Alternative Question)

Demonstrate Array Adapter using List View to display list of fruits. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ListView
        android:id="@+id/listview"
        android:layout_width="409dp"
        android:layout_height="729dp"
        app:layout_constraintBottom_toBottomOf="parent"
```

```
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

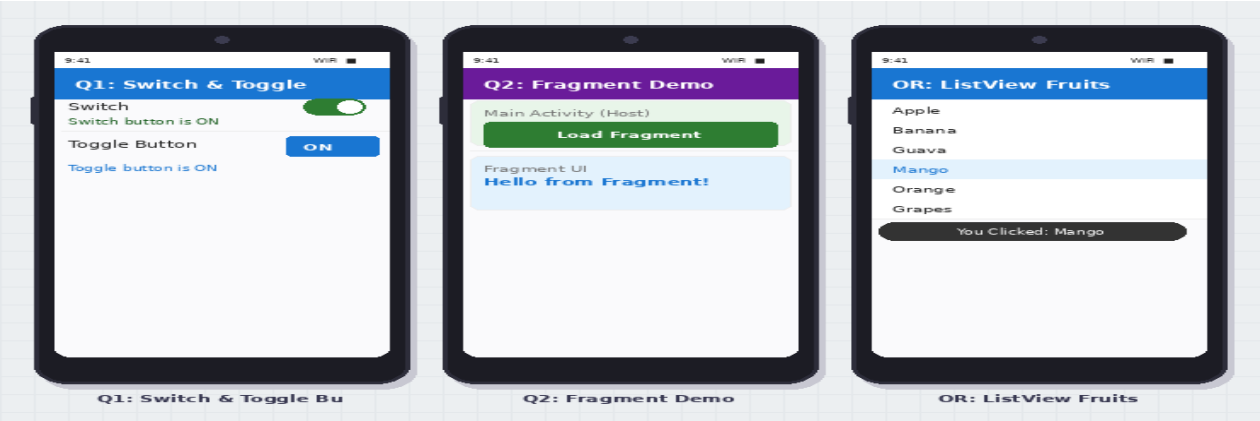
MainActivity.java

```
package com.example.listview;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    ListView l;
    String tutorials[] = { "Apple", "Banana", "Guava", "Mango", "Orange", "Grapes", "Watermelon",
"Pineapple", "Papaya" };

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        l = findViewById(R.id.listview); //support_simple_spinner_dropdown_item
        ArrayAdapter<String> arr= new ArrayAdapter<String>(this,
                androidx.appcompat.R.layout.support_simple_spinner_dropdown_item,tutorials);
        l.setAdapter(arr);

        l.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position,
                                    long id) {
                // Display a toast message with the clicked item text
                Toast.makeText(MainActivity.this, " You Clicked: " +
                        tutorials[position], Toast.LENGTH_SHORT).show();
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
ListView shows: Apple, Banana, Guava, Mango, Orange, Grapes...
Click "Mango" → Toast: "You Clicked: Mango"
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)



Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 10

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create android application to change Font Size, Color and Font Family of String. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center">

        <TextView
            android:id="@+id/textView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Font Changes"
            android:textSize="34sp" />
    </TableRow>

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent" >

        <TextView
            android:id="@+id/textView2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Enter Font Size"
            android:textSize="24sp" />

        <EditText
            android:id="@+id/editTextText"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:ems="10"
            android:inputType="text"
            android:textSize="24sp" />
    </TableRow>

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center">

        <Button
            android:id="@+id/button"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Submit"
            android:textSize="34sp" />
    </TableRow>
```

```
    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center">

        <TextView
            android:id="@+id/textView3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="MIT ACSC"
            android:textSize="24sp" />
    </TableRow>
</TableLayout>
```

MainActivity.java

```
package com.example.fontchanges;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText e;
    Button b;
    TextView t;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        e=(EditText) findViewById(R.id.editTextText);
        t=(TextView) findViewById(R.id.textView3);
        b=(Button) findViewById(R.id.button);

        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int f=Integer.parseInt(e.getText().toString());
                t.setTextSize(f);
                t.setTextColor(Color.parseColor("#00FFFF"));
            }
        });


        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Enter font size (e.g., 30) → Click Submit.
TextView "MIT ACSC" changes font size and color (Cyan) dynamically.
```

Q2 — Question

Create First Activity to accept information like Student First Name, Middle Name, Last Name, Date of birth, Address, Email ID and display all information on Second Activity when user clicks on the Submit button. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">
    <EditText android:id="@+id/etFirst"  android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="First Name"/>
    <EditText android:id="@+id/etMiddle" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Middle Name"/>
    <EditText android:id="@+id/etLast"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Last Name"/>
    <EditText android:id="@+id/etDate of Birth" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Salary" android:inputType="numberDecimal"/>
    <EditText android:id="@+id/etAddr"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Address"/>
    <EditText android:id="@+id/etEmail"  android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Email ID"/>
    <Button android:id="@+id/btnSubmit"  android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Submit"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.studentinfo;

import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText etFirst  = findViewById(R.id.etFirst);
        EditText etMiddle = findViewById(R.id.etMiddle);
        EditText etLast   = findViewById(R.id.etLast);
        EditText etDOB = findViewById(R.id.etSalary);
        EditText etAddr   = findViewById(R.id.etAddr);
        EditText etEmail  = findViewById(R.id.etEmail);

        findViewById(R.id.btnSubmit).setOnClickListener(v -> {
            Intent i = new Intent(this, MainActivity2.class);
            i.putExtra("first",  etFirst.getText().toString());
            i.putExtra("middle", etMiddle.getText().toString());
            i.putExtra("last",   etLast.getText().toString());
            i.putExtra("dob", etDOB.getText().toString());
            i.putExtra("addr",   etAddr.getText().toString());
            i.putExtra("email",  etEmail.getText().toString());
            startActivity(i);
        });
```

```
    }
}
```

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">
    <TextView android:id="@+id/tvDetails"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:textSize="16sp" android:padding="12dp" android:background="#E3F2FD"/>
</LinearLayout>
```

MainActivity2.java

```
package com.example.studentinfo;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        android.content.Intent i = getIntent();
        String info = "First Name : " + i.getStringExtra("first") +
            "\nMiddle Name: " + i.getStringExtra("middle") +
            "\nLast Name  : " + i.getStringExtra("last") +
            "\nDate of Birth     : " + i.getStringExtra("dob") +
            "\nAddress    : " + i.getStringExtra("addr") +
            "\nEmail ID   : " + i.getStringExtra("email");
        ((TextView) findViewById(R.id.tvDetails)).setText(info);
    }
}
```

Expected Output:

```
Fill all fields → Click Submit.
Activity 2 displays: First Name, Middle Name, Last Name, DOB, Address, Email ID.
```

OR

Q2 — OR (Alternative Question)

Create new contact for designing following layout (Photo, Name, Contact Number, Email). **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">

    <EditText android:id="@+id/etFirstName" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="First Name"/>
    <EditText android:id="@+id/etLastName" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Last Name"/>
    <EditText android:id="@+id/etPhone" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Phone" android:inputType="phone"/>
    <EditText android:id="@+id/etEmail" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Email"
android:inputType="textEmailAddress"/>
    <Button android:id="@+id/btnSave" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Save Contact"
        android:layout_marginTop="16dp"/>
    <TextView android:id="@+id/tvContact" android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content" android:padding="12dp"
        android:textSize="16sp" android:layout_marginTop="10dp"
        android:background="#E3F2FD"/>
</LinearLayout>
```

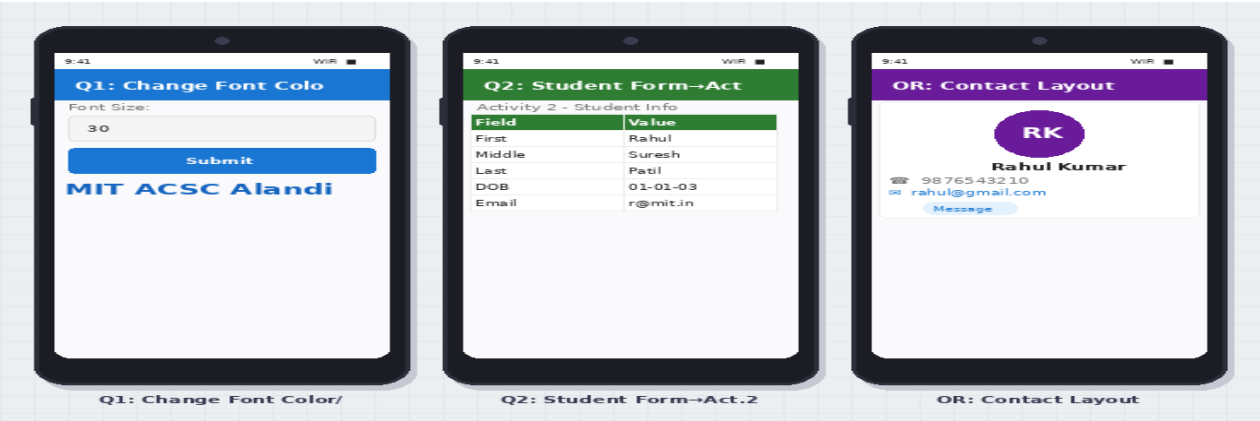
MainActivity.java

```java
package com.example.contactlayout;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText etFirst = findViewById(R.id.etFirstName);
        EditText etLast  = findViewById(R.id.etLastName);
        EditText etPhone = findViewById(R.id.etPhone);
        EditText etEmail = findViewById(R.id.etEmail);
        TextView tvContact = findViewById(R.id.tvContact);

        findViewById(R.id.btnSave).setOnClickListener(v -> {
            String details = "Name: " + etFirst.getText() + " " + etLast.getText() +
                "\nPhone: " + etPhone.getText() +
                "\nEmail: " + etEmail.getText();
            tvContact.setText(details);
            Toast.makeText(this, "Contact Saved!", Toast.LENGTH_SHORT).show();
        });
    }
}
```

Expected Output:

```
Fill in name, phone, email → Click "Save Contact".
TextView shows all contact details.
Toast: "Contact Saved!"
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)



Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 11

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create a Simple Application Which Sends "Hi" message from one activity to another with help of Button (Use Intent). **[10 Marks]**

Program Code:

activity_main.xml (Activity 1)

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Hello!"
        android:textSize="20sp"/>
</LinearLayout>
```

MainActivity.java (Activity 1)

```java
package com.example.intenthi;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button btn = findViewById(R.id.btnSend);
        btn.setOnClickListener(v -> {
            Intent i = new Intent(this, MainActivity2.class);
            i.putExtra("message", "Hi");
            startActivity(i);
        });
    }
}
```

activity_main2.xml (Activity 2)

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <TextView
        android:id="@+id/tvMsg"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="48sp"
        android:textStyle="bold"
        android:textColor="#1565C0"/>
</LinearLayout>
```

MainActivity2.java (Activity 2)

```java
package com.example.intenthi;

import android.os.Bundle;
```

```
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        TextView tv = findViewById(R.id.tvMsg);
        tv.setText(getIntent().getStringExtra("message"));
    }
}
```

Expected Output:

```
Activity 1: Click "Send Hi!" button.
Activity 2 opens and displays "Hi" in large text.
```

Q2 — Question

Create a custom "Contact" layout to hold multiple pieces of information, including: Photo, Name, Contact Number, E-mail id. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">
    <ImageView android:id="@+id/ivPhoto" android:layout_width="80dp"
        android:layout_height="80dp" android:src="@drawable/ic_person"
        android:layout_gravity="center" android:scaleType="centerCrop"/>
    <EditText android:id="@+id/etName" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Full Name"
android:layout_marginTop="12dp"/>
    <EditText android:id="@+id/etPhone" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Contact Number"
android:inputType="phone"/>
    <EditText android:id="@+id/etEmail" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="E-mail ID"
android:inputType="textEmailAddress"/>
    <Button android:id="@+id/btnSave" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Save" android:layout_gravity="center"
        android:layout_marginTop="16dp"/>
    <TextView android:id="@+id/tvResult" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:textSize="14sp" android:padding="8dp"
        android:background="#F3E5F5" android:layout_marginTop="12dp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.customcontact;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText etName  = findViewById(R.id.etName);
        EditText etPhone = findViewById(R.id.etPhone);
        EditText etEmail = findViewById(R.id.etEmail);
```

```
        TextView tvResult = findViewById(R.id.tvResult);

        findViewById(R.id.btnSave).setOnClickListener(v -> {
            String result = "Name:  " + etName.getText() +
                "\nPhone: " + etPhone.getText() +
                "\nEmail: " + etEmail.getText();
            tvResult.setText(result);
            Toast.makeText(this, "Contact Saved!", Toast.LENGTH_SHORT).show();
        });
    }
}
```

Expected Output:

```
Enter Name, Phone, Email → Click "Save".
Details displayed in a styled card below:
  Name:  Jane Smith
  Phone: 9998887776
  Email: jane@gmail.com
Toast: "Contact Saved!"
```

OR

Q2 — OR (Alternative Question)

Create an application to demonstrate date and time picker. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="221dp"
        android:layout_height="67dp"
        android:text="DatePicker"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.046" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="267dp"
        android:layout_height="57dp"
        android:text="Date"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.132" />

    <Button
        android:id="@+id/button"
```

```
        android:layout_width="199dp"
        android:layout_height="70dp"
        android:text="Date"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView3"
        app:layout_constraintVertical_bias="0.374" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.datepicker;

import android.app.DatePickerDialog;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.DatePicker;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;

//import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
    Button b;
    TextView tv;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        b=(Button) findViewById(R.id.button);
        tv=(TextView) findViewById(R.id.textView3);

        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                final Calendar c = Calendar.getInstance();

                // on below line we are getting our day, month and year.
                int year = c.get(Calendar.YEAR);
                int month = c.get(Calendar.MONTH);
                int day = c.get(Calendar.DAY_OF_MONTH);

                // on below line we are creating a variable for date picker dialog.
                DatePickerDialog datePickerDialog = new DatePickerDialog(
                        // on below line we are passing context.
                        MainActivity.this,
                        new DatePickerDialog.OnDateSetListener() {
                            @Override
                            public void onDateSet(DatePicker view, int year,
                                                  int monthOfYear, int dayOfMonth) {
                                // on below line we are setting date to our text view.
```

```
                                tv.setText(dayOfMonth + "-" + (monthOfYear + 1) + "-" + year);
                            }
                        },
                        // on below line we are passing year,month and day for selected date in our
date picker.
                        year, month, day);
                // at last we are calling show to display our date picker dialog.
                datePickerDialog.show();
            }
        });




        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="221dp"
        android:layout_height="67dp"
        android:text="TimePicker"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.046" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="267dp"
        android:layout_height="57dp"
        android:text="Time"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.132" />

    <Button
        android:id="@+id/button"
        android:layout_width="199dp"
        android:layout_height="70dp"
        android:text="Time"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
```

```
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView3"
        app:layout_constraintVertical_bias="0.374" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.timepicker;

import android.app.TimePickerDialog;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.TimePicker;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
    Button b;
    TextView tv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        b=(Button) findViewById(R.id.button);
        tv=(TextView) findViewById(R.id.textView3);

        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v)
            {
                final Calendar c = Calendar.getInstance();
                // on below line we are getting our hour, minute.
                int hour = c.get(Calendar.HOUR_OF_DAY);
                int minute = c.get(Calendar.MINUTE);

                // on below line we are initializing our Time Picker Dialog
                TimePickerDialog timePickerDialog = new TimePickerDialog(MainActivity.this,
                        new TimePickerDialog.OnTimeSetListener() {
                            @Override
                            public void onTimeSet(TimePicker view, int hourOfDay,
                                                  int minute) {
                                // on below line we are setting selected time
                                // in our text view.
                                tv.setText(hourOfDay + ":" + minute);
                            }
                        }, hour, minute, false);
                // at last we are calling show to display our time picker dialog.
                timePickerDialog.show();

            }
        });
```

```
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Date section: Click "Date" → DatePickerDialog opens → Select date → Shows: e.g., 22-3-2026
Time section: Click "Time" → TimePickerDialog opens → Select time → Shows: e.g., 14:30
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 12

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create following Vertical Scroll View Creation in Android (multiple coloured sections). **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="8dp">
        <TextView android:layout_width="match_parent" android:layout_height="120dp"
            android:background="#FF6B6B" android:text="Section 1" android:gravity="center"
            android:textColor="#FFFFFF" android:textSize="18sp" android:textStyle="bold"
            android:layout_margin="6dp"/>
        <TextView android:layout_width="match_parent" android:layout_height="120dp"
            android:background="#4CAF50" android:text="Section 2" android:gravity="center"
            android:textColor="#FFFFFF" android:textSize="18sp" android:textStyle="bold"
            android:layout_margin="6dp"/>
        <TextView android:layout_width="match_parent" android:layout_height="120dp"
            android:background="#2196F3" android:text="Section 3" android:gravity="center"
            android:textColor="#FFFFFF" android:textSize="18sp" android:textStyle="bold"
            android:layout_margin="6dp"/>
        <TextView android:layout_width="match_parent" android:layout_height="120dp"
            android:background="#FFC107" android:text="Section 4" android:gravity="center"
            android:textColor="#000000" android:textSize="18sp" android:textStyle="bold"
            android:layout_margin="6dp"/>
        <TextView android:layout_width="match_parent" android:layout_height="120dp"
            android:background="#9C27B0" android:text="Section 5" android:gravity="center"
            android:textColor="#FFFFFF" android:textSize="18sp" android:textStyle="bold"
            android:layout_margin="6dp"/>
    </LinearLayout>
</ScrollView>
```

MainActivity.java

```java
package com.example.scrollviewdemo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // ScrollView handles scrolling automatically — no Java code needed
    }
}
```

Expected Output:

```
Screen shows 5 coloured sections (Red, Green, Blue, Yellow, Purple) in a vertical ScrollView.
User can scroll up/down to view all sections.
```

Q2 — Question

Write a program to search a specific location on Google Map. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"/>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
    <uses-permission android:name="android.permission.INTERNET" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.GoogleMap"
        tools:targetApi="31">

        <!--
             TODO: Before you run your application, you need a Google Maps API key.

             To get one, follow the directions here:

                https://developers.google.com/maps/documentation/android-sdk/get-api-key

             Once you have your API key (it starts with "AIza"), define a new property in your
             project's local.properties file (e.g. MAPS_API_KEY=Aiza...), and replace the
             "YOUR_API_KEY" string in this file with "${MAPS_API_KEY}".
        -->
        <uses-library
            android:name="org.apache.http.legacy"
            android:required="false" />
        <meta-data
            android:name="com.google.android.geo.API_KEY"
            android:value="AIzaSyD4Ct8f2huj6R7ZNcrrrJrBhQjPcDeZFk0" />

        <activity
            android:name=".MapsActivity"
            android:exported="true"
            android:label="@string/title_activity_maps">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".MainActivity"
            android:exported="true">
```

```
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

```
package com.example.googlemap;

import androidx.fragment.app.FragmentActivity;

import android.os.Bundle;

import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import com.example.googlemap.databinding.ActivityMapsBinding;

public class MapsActivity extends FragmentActivity implements OnMapReadyCallback
{
    private GoogleMap mMap;
    private ActivityMapsBinding binding;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        binding = ActivityMapsBinding.inflate(getLayoutInflater());
        setContentView(binding.getRoot());

        // Obtain the SupportMapFragment and get notified when the map is ready to be used.
        SupportMapFragment mapFragment = (SupportMapFragment) getSupportFragmentManager()
                .findFragmentById(R.id.map);
        mapFragment.getMapAsync(this);
    }

    /**
     * Manipulates the map once available.
     * This callback is triggered when the map is ready to be used.
     * This is where we can add markers or lines, add listeners or move the camera. In this case,
     * we just add a marker near Alandi.
     * If Google Play services is not installed on the device, the user will be prompted to install
     * it inside the SupportMapFragment. This method will only be triggered once the user has
     * installed Google Play services and returned to the app.
     */
    @Override
    public void onMapReady(GoogleMap googleMap) {
        mMap = googleMap;

        // Add a marker in Alandi and move the camera
        // Implicit Intent to open Google Maps with searched location
// (Alternative: Use SearchOnGoogleMap approach)
LatLng searchLoc = new LatLng(18.5204, 73.8567);
mMap.addMarker(new MarkerOptions().position(searchLoc).title("Pune"));
mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(searchLoc, 14));
// Or use Intent to launch Google Maps app:
```

```
// Uri gmmIntentUri = Uri.parse("geo:0,0?q=Pune");
// Intent mapIntent = new Intent(Intent.ACTION_VIEW, gmmIntentUri);
// mapIntent.setPackage("com.google.android.apps.maps");
// startActivity(mapIntent);
    }
}
```

Expected Output:

```
App opens Google Maps → Map shows searched location (Pune) with marker.
Camera zooms to zoom level 14.
```

OR

Q2 — OR (Alternative Question)

Write an application to accept a teacher name from user and display the names of students along with subjects to whom they are teaching. Create tables: Student (sno, s_name, s_class, s_addr), Teacher (tno, t_name, qualification, experience). Student-Teacher has Many-to-Many relationship. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">

    <EditText android:id="@+id/etFirstName" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="First Name"/>
    <EditText android:id="@+id/etLastName" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Last Name"/>
    <EditText android:id="@+id/etPhone" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Phone" android:inputType="phone"/>
    <EditText android:id="@+id/etEmail" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Email"
android:inputType="textEmailAddress"/>
    <Button android:id="@+id/btnFind" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Find Students"
        android:layout_marginTop="16dp"/>
    <TextView android:id="@+id/tvContact" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:padding="12dp"
        android:textSize="16sp" android:layout_marginTop="10dp"
        android:background="#E3F2FD"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.teacherstudent;

import android.content.ContentValues;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    SQLiteDatabase db;
    EditText etTeacher;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        db = openOrCreateDatabase("SchoolDB", MODE_PRIVATE, null);
```

```
        db.execSQL("CREATE TABLE IF NOT EXISTS Student (sno INTEGER PRIMARY KEY, s_name TEXT,
s_class TEXT, s_addr TEXT)");
        db.execSQL("CREATE TABLE IF NOT EXISTS Teacher (tno INTEGER PRIMARY KEY, t_name TEXT,
qualification TEXT, experience TEXT)");
        db.execSQL("CREATE TABLE IF NOT EXISTS ST_Map (sno INTEGER, tno INTEGER, subject TEXT)");
        // Sample data
        seedData();

        etTeacher = findViewById(R.id.etTeacher);
        tvResult  = findViewById(R.id.tvResult);
        findViewById(R.id.btnFind).setOnClickListener(v -> findStudents());
    }

    private void seedData() {
        db.execSQL("INSERT OR IGNORE INTO Teacher VALUES (1,'Prof. Sharma','MCA','5 yrs')");
        db.execSQL("INSERT OR IGNORE INTO Student VALUES (1,'Rahul','TY BCA','Pune')");
        db.execSQL("INSERT OR IGNORE INTO Student VALUES (2,'Priya','TY BCA','Pune')");
        db.execSQL("INSERT OR IGNORE INTO ST_Map VALUES (1,1,'Java')");
        db.execSQL("INSERT OR IGNORE INTO ST_Map VALUES (2,1,'Android')");
    }

    private void findStudents() {
        String tname = etTeacher.getText().toString();
        String sql = "SELECT s.s_name, m.subject FROM Student s " +
            "JOIN ST_Map m ON s.sno=m.sno " +
            "JOIN Teacher t ON t.tno=m.tno WHERE t.t_name=?";
        Cursor c = db.rawQuery(sql, new String[]{tname});
        StringBuilder sb = new StringBuilder("Students taught by " + tname + ":\n");
        while (c.moveToNext())
            sb.append(c.getString(0)).append(" → ").append(c.getString(1)).append("\n");
        c.close();
        tvResult.setText(sb.length() > 20 ? sb : "No records found.");
    }
}
```

Expected Output:

```
Enter "Prof. Sharma" → Click "Find Students".
Shows: Students taught by Prof. Sharma:
  Rahul → Java
  Priya → Android
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 13

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create a Simple Application which shows Life Cycle of Activity. {Use log}. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.activitylifecycle;

import android.util.Log;

import android.os.Bundle;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Toast.makeText(getApplicationContext(), "onCreate Called", Toast.LENGTH_LONG).show();
        Log.d("LifeCycle", "onCreate Called");

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
    @Override
    protected void onStart() {
```

```
        super.onStart();
        Toast.makeText(getApplicationContext(), "onStart Called", Toast.LENGTH_LONG).show();
        Log.d("LifeCycle", "onStart Called");
    }
    @Override
    protected void onRestart() {
        super.onRestart();
        Toast.makeText(getApplicationContext(), "onRestart Called", Toast.LENGTH_LONG).show();
        Log.d("LifeCycle", "onRestart Called");
    }
    @Override
    protected void onPause() {
        super.onPause();
        Toast.makeText(getApplicationContext(), "onPause Called", Toast.LENGTH_LONG).show();
        Log.d("LifeCycle", "onPause Called");
    }
     @Override
    protected void onResume() {
        super.onResume();
        Toast.makeText(getApplicationContext(), "onResume Called", Toast.LENGTH_LONG).show();
        Log.d("LifeCycle", "onResume Called");
    }
    @Override
    protected void onStop() {
        super.onStop();
        Toast.makeText(getApplicationContext(), "onStop Called", Toast.LENGTH_LONG).show();
        Log.d("LifeCycle", "onStop Called");
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
       Toast.makeText(getApplicationContext(), "onDestroy Called", Toast.LENGTH_LONG).show();
        Log.d("LifeCycle", "onDestroy Called");
    }

}
```

Expected Output:

```
On launch: Toast messages + Logcat entries:
  D/LifeCycle: onCreate Called
  D/LifeCycle: onStart Called
  D/LifeCycle: onResume Called
On Back press:
  D/LifeCycle: onPause Called
  D/LifeCycle: onStop Called
  D/LifeCycle: onDestroy Called
```

Q2 — Question

Create the following layout which is changing android spinner text size with styles. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
    <Spinner
        android:id="@+id/spinner"
        android:layout_width="334dp"
        android:layout_height="105dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.spinner;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    String[] courses = {
            "C", "Data structures",
            "Interview prep", "Algorithms",
            "Java", "OS"
    };
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Spinner spin = findViewById(R.id.spinner);
        ArrayAdapter<String> ad = new ArrayAdapter<>(
                this,android.R.layout.simple_spinner_dropdown_item,
                courses);
        spin.setAdapter(ad);
       spin.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
           @Override
           public void onItemSelected(AdapterView<?> adapterView, View view, int i, long l) {
               Toast.makeText(MainActivity.this, " You Clicked: " +
                       courses[i], Toast.LENGTH_SHORT).show();
           }
           @Override
           public void onNothingSelected(AdapterView<?> adapterView) {

           }
       });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Spinner shows courses: C, Data Structures, Java, OS...
Select an item → Toast: "You Clicked: Java"
```

OR

Q2 — OR (Alternative Question)

Create an Android application to send email. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="129dp"
        android:layout_height="49dp"
        android:text="Email"
        android:textColor="#D11B1B"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.507"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.049" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="88dp"
        android:layout_height="35dp"
        android:text="To"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.47"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.13" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="285dp"
        android:layout_height="52dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.532"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.204" />

    <TextView
        android:id="@+id/textView3"
```

```
        android:layout_width="126dp"
        android:layout_height="40dp"
        android:text="Subject"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.29" />

    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="283dp"
        android:layout_height="51dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.532"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.379" />

    <TextView
        android:id="@+id/textView4"
        android:layout_width="157dp"
        android:layout_height="40dp"
        android:text="Body"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />

    <EditText
        android:id="@+id/editTextText3"
        android:layout_width="278dp"
        android:layout_height="54dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.532"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.618" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Email"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.505"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.813" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

AndroidManifest.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.INTERNET" />
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Email"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

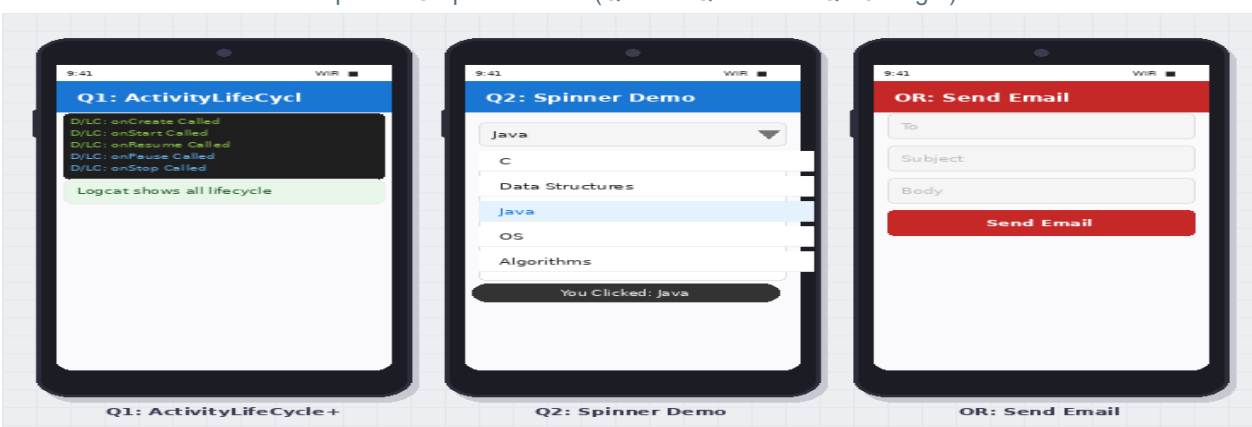
```java
package com.example.email;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Button send;
        EditText to, subject, body;
        to=(EditText) findViewById(R.id.editTextText);
        subject=(EditText) findViewById(R.id.editTextText2);
        body=(EditText) findViewById(R.id.editTextText3);
        send=(Button) findViewById(R.id.button);
        send.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String emailto = to.getText().toString();
                String emailsubject = subject.getText().toString();
```

```
                String emailbody = body.getText().toString();
                // define Intent object with action attribute as ACTION_SEND
                Intent intent = new Intent(Intent.ACTION_SEND);
                // add three fields to intent using putExtra function
                intent.putExtra(Intent.EXTRA_EMAIL, new String[]{emailto});
                intent.putExtra(Intent.EXTRA_SUBJECT, emailsubject);
                intent.putExtra(Intent.EXTRA_TEXT, emailbody);
                // set type of intent
                intent.setType("message/rfc822");
  // startActivity with intent with chooser as Email client using createChooser function
                startActivity(Intent.createChooser(intent, "Choose an Email client :"));
            }
        });


        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Fill To, Subject, Body → Click "Send Email".
Email chooser opens → Gmail/Outlook opens with fields pre-filled.
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)



Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 14

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Design following — add a border to an Android Layout. **[10 Marks]**

Program Code:

res/drawable/border_bg.xml (Shape Drawable)

```xml
<?xml version="1.0" encoding="utf-8"?>
<!-- res/drawable/border_bg.xml -->
<!-- Create this file in res/drawable/ -->
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#FFFFFF"/>
    <stroke android:width="3dp" android:color="#1565C0"/>
    <corners android:radius="8dp"/>
    <padding android:left="12dp" android:top="12dp"
             android:right="12dp" android:bottom="12dp"/>
</shape>
```

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:padding="20dp" android:gravity="center">
    <LinearLayout
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:orientation="vertical"
        android:background="@drawable/border_bg"
        android:gravity="center" android:padding="16dp">
        <TextView android:layout_width="wrap_content" android:layout_height="wrap_content"
            android:text="Bordered Layout" android:textSize="22sp"
            android:textColor="#1565C0" android:textStyle="bold"/>
        <TextView android:layout_width="wrap_content" android:layout_height="wrap_content"
            android:text="MIT ACSC, Alandi" android:textSize="16sp"
            android:textColor="#546E7A" android:layout_marginTop="8dp"/>
    </LinearLayout>
</RelativeLayout>
```

MainActivity.java

```java
package com.example.borderlayout;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Border is applied via drawable background — no extra Java needed
    }
}
```

Expected Output:

```
Screen shows a box with a blue border (3dp, rounded corners).
Inside: "Bordered Layout" heading and "MIT ACSC, Alandi" subtitle.
```

Q2 — Question

Create simple application with Login Screen. On successful login, give message and go to next Activity (Without Using Database). **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="183dp"
        android:layout_height="65dp"
        android:text="Login"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.674"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.058" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="135dp"
        android:layout_height="26dp"
        android:layout_marginBottom="336dp"
        android:text="Username"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.445"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.081" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.417"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView2"
        app:layout_constraintVertical_bias="0.098" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="196dp"
        android:layout_height="46dp"
        android:text="Password"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText"
        app:layout_constraintVertical_bias="0.095" />
```

```
    <Button
        android:id="@+id/button"
        android:layout_width="122dp"
        android:layout_height="76dp"
        android:layout_marginBottom="76dp"
        android:text="Login"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.47"
        app:layout_constraintStart_toStartOf="parent" />

    <EditText
        android:id="@+id/editTextTextPassword"
        android:layout_width="211dp"
        android:layout_height="54dp"
        android:layout_marginBottom="140dp"
        android:ems="10"
        android:inputType="textPassword"
        app:layout_constraintBottom_toTopOf="@+id/button"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.381"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView3"
        app:layout_constraintVertical_bias="1.0" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">

    <TextView
        android:id="@+id/textView5"
        android:layout_width="248dp"
        android:layout_height="81dp"
        android:text="Login Message"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.711"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.12" />

    <TextView
        android:id="@+id/textView6"
        android:layout_width="321dp"
        android:layout_height="83dp"
        android:layout_marginBottom="300dp"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.602"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView5"
```

```
        app:layout_constraintVertical_bias="0.394" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.loginonnextactivity;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText a,b;
    Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        a = (EditText) findViewById(R.id.editTextText);
        b = (EditText) findViewById(R.id.editTextTextPassword);
        btn = (Button) findViewById(R.id.button);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String aaa = a.getText().toString();
                String bbb = b.getText().toString();

                Intent i = new Intent(MainActivity.this, MainActivity2.class);
                i.putExtra("uname", aaa);
                i.putExtra("password", bbb);
                startActivity(i);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

MainActivity2.java

```
package com.example.loginonnextactivity;

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity2 extends AppCompatActivity {
    TextView v;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main2);
        v=findViewById(R.id.textView6);
        Intent i=getIntent();
        String a=i.getStringExtra("uname");
        String b= i.getStringExtra("password");
        if(a.equals("Admin")&&b.equals("123"))
         v.setText(String.valueOf("Welcome Admin"));
        else
            v.setText(String.valueOf("Invalid Login"));

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Username: Admin, Password: 123 → Login.
Activity 2: "Welcome Admin"
Wrong credentials → "Invalid Login"
```

OR

Q2 — OR (Alternative Question)

Create First Activity to accept information like Employee First Name, Middle Name, Last Name, Salary, Address, Email ID and display all information on Second Activity when user clicks on Submit button. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">
    <EditText android:id="@+id/etFirst"  android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="First Name"/>
    <EditText android:id="@+id/etMiddle" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Middle Name"/>
    <EditText android:id="@+id/etLast"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Last Name"/>
    <EditText android:id="@+id/etSalary" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Salary" android:inputType="numberDecimal"/>
    <EditText android:id="@+id/etAddr"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Address"/>
    <EditText android:id="@+id/etEmail"  android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Email ID"/>
    <Button android:id="@+id/btnSubmit"  android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Submit"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.employeeform;

import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
```

```java
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText etFirst  = findViewById(R.id.etFirst);
        EditText etMiddle = findViewById(R.id.etMiddle);
        EditText etLast   = findViewById(R.id.etLast);
        EditText etSalary = findViewById(R.id.etSalary);
        EditText etAddr   = findViewById(R.id.etAddr);
        EditText etEmail  = findViewById(R.id.etEmail);

        findViewById(R.id.btnSubmit).setOnClickListener(v -> {
            Intent i = new Intent(this, MainActivity2.class);
            i.putExtra("first",  etFirst.getText().toString());
            i.putExtra("middle", etMiddle.getText().toString());
            i.putExtra("last",   etLast.getText().toString());
            i.putExtra("salary", etSalary.getText().toString());
            i.putExtra("addr",   etAddr.getText().toString());
            i.putExtra("email",  etEmail.getText().toString);
            startActivity(i);
        });
    }
}
```

activity_main2.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">
    <TextView android:id="@+id/tvDetails"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:textSize="16sp" android:padding="12dp" android:background="#E3F2FD"/>
</LinearLayout>
```

MainActivity2.java

```java
package com.example.employeeform;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        android.content.Intent i = getIntent();
        String info = "First Name : " + i.getStringExtra("first") +
            "\nMiddle Name: " + i.getStringExtra("middle") +
            "\nLast Name  : " + i.getStringExtra("last") +
            "\nSalary     : " + i.getStringExtra("salary") +
            "\nAddress    : " + i.getStringExtra("addr") +
            "\nEmail ID   : " + i.getStringExtra("email");
        ((TextView) findViewById(R.id.tvDetails)).setText(info);
    }
}
```

Expected Output:

```
Fill all employee fields → Submit.
Activity 2 shows: First Name, Middle Name, Last Name, Salary, Address, Email.
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 15

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create an Android App — it reads the Students Details (Name, Surname, Class, Gender, Hobbies, Marks) and display all information in another activity in table format on click of Submit button. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="16dp">
    <EditText android:id="@+id/etName"    android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Name"/>
    <EditText android:id="@+id/etSurname" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Surname"/>
    <EditText android:id="@+id/etClass"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Class"/>
    <RadioGroup android:id="@+id/rgGender" android:layout_width="match_parent"
android:layout_height="wrap_content" android:orientation="horizontal">
        <RadioButton android:id="@+id/rbMale"   android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Male" android:checked="true"/>
        <RadioButton android:id="@+id/rbFemale" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Female"/>
    </RadioGroup>
    <EditText android:id="@+id/etHobbies" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Hobbies"/>
    <EditText android:id="@+id/etMarks"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Marks" android:inputType="number"/>
    <Button android:id="@+id/btnSubmit"   android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Submit" android:layout_marginTop="12dp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.studentdetails;

import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText etName    = findViewById(R.id.etName);
        EditText etSurname = findViewById(R.id.etSurname);
        EditText etClass   = findViewById(R.id.etClass);
        RadioGroup rgGender = findViewById(R.id.rgGender);
        EditText etHobbies = findViewById(R.id.etHobbies);
        EditText etMarks   = findViewById(R.id.etMarks);

        findViewById(R.id.btnSubmit).setOnClickListener(v -> {
            RadioButton rb = findViewById(rgGender.getCheckedRadioButtonId());
            Intent i = new Intent(this, MainActivity2.class);
            i.putExtra("name",    etName.getText().toString());
            i.putExtra("surname", etSurname.getText().toString());
            i.putExtra("clazz",   etClass.getText().toString());
            i.putExtra("gender",  rb.getText().toString());
            i.putExtra("hobbies", etHobbies.getText().toString());
```

```
            i.putExtra("marks",   etMarks.getText().toString());
            startActivity(i);
        });
    }
}
```

activity_main2.xml (TableLayout)

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
android:padding="16dp">
    <TableLayout android:id="@+id/tableLayout"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:stretchColumns="1" android:shrinkColumns="1">
    </TableLayout>
</ScrollView>
```

MainActivity2.java

```
package com.example.studentdetails;

import android.os.Bundle;
import android.view.Gravity;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        android.content.Intent intent = getIntent();
        TableLayout tl = findViewById(R.id.tableLayout);
        String[][] data = {
            {"Field","Value"},
            {"Name",    intent.getStringExtra("name")},
            {"Surname", intent.getStringExtra("surname")},
            {"Class",   intent.getStringExtra("clazz")},
            {"Gender",  intent.getStringExtra("gender")},
            {"Hobbies", intent.getStringExtra("hobbies")},
            {"Marks",   intent.getStringExtra("marks")},
        };
        boolean header = true;
        for (String[] row : data) {
            TableRow tr = new TableRow(this);
            for (String cell : row) {
                TextView tv = new TextView(this);
                tv.setText(cell);
                tv.setPadding(12,8,12,8);
                tv.setTextSize(15);
                if (header) { tv.setTypeface(null, android.graphics.Typeface.BOLD);
                    tv.setBackgroundColor(0xFF1565C0); tv.setTextColor(0xFFFFFFFF); }
                else { tv.setBackgroundColor(0xFFE3F2FD); tv.setTextColor(0xFF212121); }
                tr.addView(tv);
            }
            tl.addView(tr);
            header = false;
        }
    }
}
```

Expected Output:

```
Activity 1: Fill all student fields → Submit.
Activity 2: Displays a table with all student details (header in blue).
```

Q2 — Question

Create an Android Application that Demonstrate TimePicker and display Selected Time on TextView. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="221dp"
        android:layout_height="67dp"
        android:text="TimePicker"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.046" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="267dp"
        android:layout_height="57dp"
        android:text="Time"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.132" />

    <Button
        android:id="@+id/button"
        android:layout_width="199dp"
        android:layout_height="70dp"
        android:text="Time"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView3"
        app:layout_constraintVertical_bias="0.374" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.timepicker;

import android.app.TimePickerDialog;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
```

```java
import android.widget.TimePicker;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
    Button b;
    TextView tv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        b=(Button) findViewById(R.id.button);
        tv=(TextView) findViewById(R.id.textView3);

        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v)
            {
                final Calendar c = Calendar.getInstance();
                // on below line we are getting our hour, minute.
                int hour = c.get(Calendar.HOUR_OF_DAY);
                int minute = c.get(Calendar.MINUTE);

                // on below line we are initializing our Time Picker Dialog
                TimePickerDialog timePickerDialog = new TimePickerDialog(MainActivity.this,
                        new TimePickerDialog.OnTimeSetListener() {
                            @Override
                            public void onTimeSet(TimePicker view, int hourOfDay,
                                                  int minute) {
                                // on below line we are setting selected time
                                // in our text view.
                                tv.setText(hourOfDay + ":" + minute);
                            }
                        }, hour, minute, false);
                // at last we are calling show to display our time picker dialog.
                timePickerDialog.show();

            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Click 'Time' button → TimePickerDialog appears.
Select 3:45 PM → TextView shows: "15:45"
```

OR

Q2 — OR (Alternative Question)

Create a Simple calculator. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Calculator"
        android:textSize="48sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.05" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="182dp"
        android:layout_height="60dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.48"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.064" />

    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="178dp"
        android:layout_height="63dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText"
        app:layout_constraintVertical_bias="0.048" />

    <Button
        android:id="@+id/button"
        android:layout_width="132dp"
        android:layout_height="87dp"
        android:layout_marginBottom="144dp"
        android:text="Add"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.471"
```

```
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText2"
        app:layout_constraintVertical_bias="0.192" />

    <Button
        android:id="@+id/button2"
        android:layout_width="135dp"
        android:layout_height="86dp"
        android:text="Sub"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.467"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/button"
        app:layout_constraintVertical_bias="0.106" />

    <Button
        android:id="@+id/button3"
        android:layout_width="126dp"
        android:layout_height="74dp"
        android:text="Mult"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/button2"
        app:layout_constraintVertical_bias="0.092" />

    <Button
        android:id="@+id/button4"
        android:layout_width="128dp"
        android:layout_height="72dp"
        android:text="Div"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/button3" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.simplecalculator;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText e1,e2;
    Button b1,b2,b3,b4;
```

```
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        e1=findViewById(R.id.editTextText);
        e2=findViewById(R.id.editTextText2);
        b1=findViewById(R.id.button);
        b2=findViewById(R.id.button2);
        b3=findViewById(R.id.button3);
        b4=findViewById(R.id.button4);



        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int a=Integer.parseInt(e1.getText().toString());
                int b=Integer.parseInt(e2.getText().toString());
                int c=a+b;
                Toast.makeText(getApplicationContext(),"Addition =
"+String.valueOf(c),Toast.LENGTH_LONG).show();
            }
        });

        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int a=Integer.parseInt(e1.getText().toString());
                int b=Integer.parseInt(e2.getText().toString());
                int c=a-b;
                Toast.makeText(getApplicationContext(),"Subtraction =
"+String.valueOf(c),Toast.LENGTH_LONG).show();
            }
        });

        b3.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int a=Integer.parseInt(e1.getText().toString());
                int b=Integer.parseInt(e2.getText().toString());
                int c=a*b;
                Toast.makeText(getApplicationContext(),"Multiplication =
"+String.valueOf(c),Toast.LENGTH_LONG).show();
            }
        });

        b4.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int a=Integer.parseInt(e1.getText().toString());
                int b=Integer.parseInt(e2.getText().toString());
                int c=a/b;
                Toast.makeText(getApplicationContext(),"Division =
"+String.valueOf(c),Toast.LENGTH_LONG).show();
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
```

```
        });
    }
}
```

Expected Output:

```
Enter two numbers → Click any operation button.
Add → Toast: "Addition = 19"
Sub → Toast: "Subtraction = 5"
Mult → Toast: "Multiplication = 84"
Div → Toast: "Division = 3"
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 16

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Write an android code to make phone call using Intent. **[10 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="318dp"
        android:layout_height="69dp"
        android:text="Phone Calling"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.827"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.078" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="282dp"
        android:layout_height="66dp"
        android:text="Enter Mobile No"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.637"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.089" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="294dp"
        android:layout_height="60dp"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.572"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView2"
        app:layout_constraintVertical_bias="0.052" />

    <Button
        android:id="@+id/button"
        android:layout_width="207dp"
        android:layout_height="73dp"
        android:text="Dial"
        android:textSize="34sp"
```

```
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.544"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText"
        app:layout_constraintVertical_bias="0.133" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-feature
        android:name="android.hardware.telephony"
        android:required="false" />
    <uses-permission android:name="android.permission.CALL_PHONE" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.PhoneCall"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

```
package com.example.phonecall;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
```

```
        EditText mno;
        Button call;
        mno = findViewById(R.id.editTextText);
        call = findViewById(R.id.button);
        call.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String phone_number = mno.getText().toString();
                 // Getting instance of Intent with action as ACTION_CALL
                Intent phone_intent = new Intent(Intent.ACTION_CALL);
                // Set data of Intent through Uri by parsing phone number
                phone_intent.setData(Uri.parse("tel:" + phone_number));
                // start Intent
                startActivity(phone_intent);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Enter mobile number → Click 'Dial'.
Phone dialer opens with number pre-filled.
Requires CALL_PHONE permission.
```

Q2 — Question

Create an android application that demonstrate Spinner. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Spinner
        android:id="@+id/spinner"
        android:layout_width="334dp"
        android:layout_height="105dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.spinner;

import android.os.Bundle;
```

```java
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    String[] courses = {
            "C", "Data structures",
            "Interview prep", "Algorithms",
            "Java", "OS"
    };
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Spinner spin = findViewById(R.id.spinner);
        ArrayAdapter<String> ad = new ArrayAdapter<>(
                this,android.R.layout.simple_spinner_dropdown_item,
                courses);
        spin.setAdapter(ad);
       spin.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
           @Override
           public void onItemSelected(AdapterView<?> adapterView, View view, int i, long l) {
               Toast.makeText(MainActivity.this, " You Clicked: " +
                       courses[i], Toast.LENGTH_SHORT).show();
           }
           @Override
           public void onNothingSelected(AdapterView<?> adapterView) {

           }
       });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Spinner shows dropdown list: C, Data Structures, Java, OS, Algorithms...
Select an item → Toast: "You Clicked: Java"
```

OR

Q2 — OR (Alternative Question)

Construct an Android Application to accept a number and calculate Factorial and Sum of Digits of a given number using Context Menu. **[20 Marks]**

Program Code:

📄 activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="20dp" android:gravity="center">
    <EditText android:id="@+id/etNumber" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Enter a number"
        android:inputType="number"/>
    <TextView android:id="@+id/tvLongPress"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Long Press here for Context Menu"
        android:textSize="16sp" android:padding="12dp"
        android:layout_marginTop="16dp" android:background="#E3F2FD"/>
    <TextView android:id="@+id/tvResult"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:textSize="18sp" android:textStyle="bold"
        android:layout_marginTop="20dp" android:textColor="#1565C0"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.contextmenufactorial;

import android.os.Bundle;
import android.view.*;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText etNumber;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        etNumber = findViewById(R.id.etNumber);
        tvResult = findViewById(R.id.tvResult);
        registerForContextMenu(findViewById(R.id.tvLongPress));
    }

    @Override
    public void onCreateContextMenu(ContextMenu menu, View v, ContextMenu.ContextMenuInfo info) {
        super.onCreateContextMenu(menu, v, info);
        menu.setHeaderTitle("Choose Operation");
        menu.add(0, 1, 0, "Factorial");
        menu.add(0, 2, 0, "Sum of Digits");
    }

    @Override
    public boolean onContextItemSelected(MenuItem item) {
        int n = Integer.parseInt(etNumber.getText().toString());
        if (item.getItemId() == 1) {
            int f = 1; for (int i = 1; i <= n; i++) f *= i;
            tvResult.setText("Factorial of " + n + " = " + f);
        } else {
            int sum = 0, temp = n;
            while (temp > 0) { sum += temp % 10; temp /= 10; }
            tvResult.setText("Sum of Digits of " + n + " = " + sum);
        }
        return true;
    }
}
```

Expected Output:

```
Enter 5 → Long press text → Context menu shows: "Factorial" and "Sum of Digits"
Select Factorial → "Factorial of 5 = 120"
Select Sum of Digits → "Sum of Digits of 5 = 5"
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 17

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create an Android Application that Demonstrate Alert Dialog Box. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button"
        android:layout_width="287dp"
        android:layout_height="117dp"
        android:text="Alert Button"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.alertdialog;

import android.annotation.SuppressLint;
import android.content.DialogInterface;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

            Button b;
            b=findViewById(R.id.button);
            b.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
            public void onClick(View v) {
                AlertDialog.Builder adb = new AlertDialog.Builder(MainActivity.this);
                adb.setTitle("Confirm Exit !");
                adb.setMessage("are you sure, you want to Exit ?");
                adb.setCancelable(true);
                adb.setPositiveButton("OK", new DialogInterface.OnClickListener() {
                    @Override
                    public void onClick(DialogInterface dialog, int which) {
                        Toast.makeText(getApplicationContext(),"You pressed
OK",Toast.LENGTH_LONG).show();
                        //finish();
                    }
                });
                adb.setNegativeButton("Cancel", new DialogInterface.OnClickListener() {
                    @Override
                    public void onClick(DialogInterface dialog, int which) {
                        Toast.makeText(getApplicationContext(),"You pressed
Cancel",Toast.LENGTH_LONG).show();
                    }
                });
                adb.setNeutralButton("Retry", new DialogInterface.OnClickListener() {
                    @Override
                    public void onClick(DialogInterface dialog, int which) {
                        Toast.makeText(getApplicationContext(),"You Presses
Retry",Toast.LENGTH_LONG).show();
                    }
                });
                // Show the dialog
                adb.create();
                adb.show();

            }
        });



        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Click 'Alert Button' → AlertDialog appears.
Title: "Confirm Exit !"
Buttons: Retry / Cancel / OK
Click OK → Toast: "You pressed OK"
Click Cancel → Toast: "You pressed Cancel"
```

Q2 — Question

Create an Android Application that produces Notification. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
```

```
    <Button
        android:id="@+id/btnNotify"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Notification"
        android:textSize="18sp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.notificationdemo;

import android.app.*;
import android.content.Intent;
import android.os.Build;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;

public class MainActivity extends AppCompatActivity {
    static final String CHANNEL_ID = "demo_channel";
    static final int NOTIF_ID = 1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        createNotificationChannel();
        Button btn = findViewById(R.id.btnNotify);
        btn.setOnClickListener(v -> sendNotification());
    }

    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            NotificationChannel channel = new NotificationChannel(
                CHANNEL_ID, "Demo Channel", NotificationManager.IMPORTANCE_DEFAULT);
            NotificationManager mgr = getSystemService(NotificationManager.class);
            mgr.createNotificationChannel(channel);
        }
    }

    private void sendNotification() {
        Intent intent = new Intent(this, MainActivity.class);
        PendingIntent pi = PendingIntent.getActivity(this, 0, intent,
            PendingIntent.FLAG_IMMUTABLE);
        NotificationCompat.Builder builder = new NotificationCompat.Builder(this, CHANNEL_ID)
            .setSmallIcon(android.R.drawable.ic_dialog_info)
            .setContentTitle("Hello!")
            .setContentText("This is a notification message.")
            .setContentIntent(pi)
            .setAutoCancel(true);
        NotificationManager mgr = (NotificationManager) getSystemService(NOTIFICATION_SERVICE);
        mgr.notify(NOTIF_ID, builder.build());
    }
}
```

Expected Output:

```
Click "Send Notification" → Notification appears in status bar.
Title: "Hello!" | Body: "This is a notification message."
Tap notification → Opens the app.
```

OR

Q2 — OR (Alternative Question)

Create an Android Application to accept two numbers and find power and Average. Display the result on the next activity using Context Menu. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="20dp" android:gravity="center">
    <EditText android:id="@+id/etBase" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Enter Base"
android:inputType="numberDecimal"/>
    <EditText android:id="@+id/etExp" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Enter Exponent"
android:inputType="numberDecimal"/>
    <TextView android:id="@+id/tvLongPress"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Long Press here for Context Menu" android:textSize="16sp"
        android:textColor="#4A148C" android:padding="12dp" android:layout_marginTop="16dp"
        android:background="#F3E5F5"/>
    <TextView android:id="@+id/tvResult"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:textSize="18sp" android:textStyle="bold" android:layout_marginTop="20dp"
        android:textColor="#1565C0"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.contextmenupoweravg;

import android.os.Bundle;
import android.view.*;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText etBase, etExp;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        etBase   = findViewById(R.id.etBase);
        etExp    = findViewById(R.id.etExp);
        tvResult = findViewById(R.id.tvResult);
        TextView tvLP = findViewById(R.id.tvLongPress);
        registerForContextMenu(tvLP);
    }

    @Override
    public void onCreateContextMenu(ContextMenu menu, View v, ContextMenu.ContextMenuInfo menuInfo)
{
        super.onCreateContextMenu(menu, v, menuInfo);
        menu.setHeaderTitle("Calculate");
        menu.add(0, 1, 0, "Power");
        menu.add(0, 2, 0, "Average");
    }

    @Override
    public boolean onContextItemSelected(MenuItem item) {
        double base = Double.parseDouble(etBase.getText().toString());
        double exp  = Double.parseDouble(etExp.getText().toString());
        if (item.getItemId() == 1) {
```

```
            tvResult.setText("Power = " + Math.pow(base, exp));
        } else {
            tvResult.setText("Average = " + (base + exp) / 2);
        }
        return true;
    }
}
```

Expected Output:

```
Enter Base: 2, Exponent: 3 → Long press text → Context menu.
Select "Power"   → "Power = 8.0"
Select "Average" → "Average = 2.5"
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 18

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create an Android Application that on/off the bulb using Toggle Button. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:gravity="center" android:padding="20dp">
    <ImageView android:id="@+id/ivBulb"
        android:layout_width="120dp" android:layout_height="120dp"
        android:src="@drawable/bulb_off"/>
    <TextView android:id="@+id/tvStatus"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Bulb is OFF" android:textSize="18sp"
        android:textColor="#B71C1C" android:layout_marginTop="12dp"/>
    <ToggleButton android:id="@+id/toggleBulb"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:textOn="Turn OFF Bulb" android:textOff="Turn ON Bulb"
        android:layout_marginTop="16dp" android:textSize="16sp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.bulbtoggle;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        ImageView ivBulb    = findViewById(R.id.ivBulb);
        TextView tvStatus   = findViewById(R.id.tvStatus);
        ToggleButton toggle = findViewById(R.id.toggleBulb);

        toggle.setOnCheckedChangeListener((btn, isChecked) -> {
            if (isChecked) {
                ivBulb.setImageResource(R.drawable.bulb_on);
                ivBulb.setBackgroundColor(0xFFFFD600);
                tvStatus.setText("Bulb is ON");
                tvStatus.setTextColor(0xFF2E7D32);
            } else {
                ivBulb.setImageResource(R.drawable.bulb_off);
                ivBulb.setBackgroundColor(0xFFBBBBBB);
                tvStatus.setText("Bulb is OFF");
                tvStatus.setTextColor(0xFFB71C1C);
            }
        });
    }
}
```

Expected Output:

```
Toggle ON → Bulb image changes to ON (yellow glow), text: "Bulb is ON" (green)
Toggle OFF → Bulb image changes to OFF (grey), text: "Bulb is OFF" (red)
Note: Add bulb_on.png and bulb_off.png to res/drawable/
```

Q2 — Question

Design Following Screens using Table Layout. Display the entered text on next activity. **[20 Marks]**

Program Code:

activity_main.xml (TableLayout)

```xml
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:padding="16dp" android:stretchColumns="1">
    <TableRow>
        <TextView android:text="Name:" android:textSize="16sp" android:padding="8dp"/>
        <EditText android:id="@+id/etName" android:layout_width="0dp" android:layout_weight="1"
            android:hint="Enter Name"/>
    </TableRow>
    <TableRow>
        <TextView android:text="Roll No:" android:textSize="16sp" android:padding="8dp"/>
        <EditText android:id="@+id/etRoll" android:layout_width="0dp" android:layout_weight="1"
            android:hint="Enter Roll No"/>
    </TableRow>
    <TableRow>
        <TextView android:text="Class:" android:textSize="16sp" android:padding="8dp"/>
        <EditText android:id="@+id/etClass" android:layout_width="0dp" android:layout_weight="1"
            android:hint="Enter Class"/>
    </TableRow>
    <TableRow android:gravity="center">
        <Button android:id="@+id/btnSubmit" android:layout_width="wrap_content"
            android:layout_height="wrap_content" android:text="Submit"
            android:layout_span="2" android:layout_marginTop="16dp"/>
    </TableRow>
</TableLayout>
```

MainActivity.java

```java
package com.example.tableformdemo;

import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText etName  = findViewById(R.id.etName);
        EditText etRoll  = findViewById(R.id.etRoll);
        EditText etClass = findViewById(R.id.etClass);

        findViewById(R.id.btnSubmit).setOnClickListener(v -> {
            Intent i = new Intent(this, MainActivity2.class);
            i.putExtra("name",  etName.getText().toString());
            i.putExtra("roll",  etRoll.getText().toString());
            i.putExtra("clazz", etClass.getText().toString());
            startActivity(i);
        });
    }
}
```

activity_main2.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:padding="20dp" android:gravity="center">
    <TextView android:id="@+id/tvDetails"
```

```
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:textSize="18sp" android:padding="16dp" android:background="#E8F5E9"
        android:textColor="#1B5E20"/>
</LinearLayout>
```

📄 **MainActivity2.java**

```
package com.example.tableformdemo;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        android.content.Intent i = getIntent();
        String info = "Name  : " + i.getStringExtra("name") +
            "\nRoll  : " + i.getStringExtra("roll") +
            "\nClass : " + i.getStringExtra("clazz");
        ((TextView) findViewById(R.id.tvDetails)).setText(info);
    }
}
```

Expected Output:

```
Activity 1 (TableLayout): Fill Name, Roll No, Class → Submit.
Activity 2 shows:
  Name  : Rahul Patil
  Roll  : 101
  Class : TY BCA
```

OR

Q2 — OR (Alternative Question)

Create application to send SMS message. After sending message display delivery report of message. **[20 Marks]**

Program Code:

📄 **activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="119dp"
        android:layout_height="44dp"
        android:text="SMS"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.061" />

    <TextView
        android:id="@+id/textView2"
```

```
        android:layout_width="217dp"
        android:layout_height="37dp"
        android:text="Enter Mobile No"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        app:layout_constraintVertical_bias="0.05" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.482"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView2"
        app:layout_constraintVertical_bias="0.026" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="228dp"
        android:layout_height="49dp"
        android:text="Enter SMS Text"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.53"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText"
        app:layout_constraintVertical_bias="0.057" />

    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.482"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView3"
        app:layout_constraint Vertical_bias="0.06" />

    <Button
        android:id="@+id/button"
        android:layout_width="228dp"
        android:layout_height="61dp"
        android:text="Send SMS"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextText2"
        app:layout_constraintVertical_bias="0.159" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-feature
        android:name="android.hardware.telephony"
        android:required="false" />
    <uses-permission android:name="android.permission.SEND_SMS"/>

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SMS"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

```
package com.example.sms;

import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        EditText phonenumber,message;
        Button send;
        send=findViewById(R.id.button);
        phonenumber=findViewById(R.id.editTextText);
        message=findViewById(R.id.editTextText2);
        send.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
```

```
                String number=phonenumber.getText().toString();
                String msg=message.getText().toString();

                SmsManager smsManager=SmsManager.getDefault();
                smsManager.sendTextMessage(number,null,msg,null,null);
                Toast.makeText(getApplicationContext(),"Message Sent",Toast.LENGTH_LONG).show();

            }
        });


        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Expected Output:

```
Enter mobile number and SMS text → Click "Send SMS".
Toast: "Message Sent"
SMS delivered to recipient device.
Requires: SEND_SMS permission in AndroidManifest.xml
Note: Works on a physical device only.
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 19

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create Android Program to Change the Image on the Screen. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageSwitcher
        android:id="@+id/imgsw"
        android:layout_width="match_parent"
        android:layout_height="600dp"
        android:layout_margin="20dp">

    </ImageSwitcher>

    <Button
        android:id="@+id/prev"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="prev"
        android:layout_below="@id/imgsw"
        >

    </Button>

    <Button
        android:id="@+id/next"
        android:layout_below="@id/imgsw"
        android:layout_alignParentRight="true"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="next">

    </Button>
</RelativeLayout>
```

MainActivity.java

```java
package com.example.imageswitchdemo;

import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.ImageSwitcher;
import android.widget.ImageView;
import android.widget.ViewSwitcher;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```java
public class MainActivity extends AppCompatActivity {

    //Integer[] images={R.drawable.img1,R.drawable.img2};
    int i=0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button next=findViewById(R.id.next);
        Button prev=findViewById(R.id.prev);
        ImageSwitcher imgsw=findViewById(R.id.imgsw);

        imgsw.setFactory(new ViewSwitcher.ViewFactory() {
            @Override
            public View makeView() {
                ImageView imageView=new ImageView(getApplicationContext());
                imageView.setScaleType(ImageView.ScaleType.FIT_CENTER);
                return imageView;
            }
        });
        //Animation out= AnimationUtils.loadAnimation(this, android.R.anim.slide_out_right);
        //Animation in= AnimationUtils.loadAnimation(this, android.R.anim.slide_in_left);
        //imgsw.setOutAnimation(out);
        //.setInAnimation(in);


        next.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                //if(i<images.length){
                  //  i++;
                    //imgsw.setImageResource(images[i]);
                imgsw.setImageResource(R.drawable.img2);
                }
        });

        prev.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                //if(i>0){
                    //i--;
                    //imgsw.setImageResource(images[i]);
                imgsw.setImageResource(R.drawable.img1);

                }




        });

    }
}
```

Expected Output:

```
App loads with first image (img1) displayed in ImageSwitcher.
Click 'next' → Shows img2.
Click 'prev' → Shows img1.
Note: Add img1.png and img2.png to res/drawable/
```

Q2 — Question

Demonstrate Options Menu, Context Menu and Popup Menu in android. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>


optionmenubar.xml

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/o1"
        android:title="BCA" />
    <item
        android:id="@+id/o2"
        android:title="BCS" />
    <item
        android:id="@+id/o3"
        android:title="BCOM" />
</menu>

gradle.properties

# Project-wide Gradle settings.
# IDE (e.g. Android Studio) users:
# Gradle settings configured through the IDE *will override*
# any settings specified in this file.
# For more details on how to configure your build environment visit
# http://www.gradle.org/docs/current/userguide/build_environment.html
# Specifies the JVM arguments used for the daemon process.
# The setting is particularly useful for tweaking memory settings.
org.gradle.jvmargs=-Xmx2048m -Dfile.encoding=UTF-8
# When configured, Gradle will run in incubating parallel mode.
# This option should only be used with decoupled projects. For more details, visit
# https://developer.android.com/r/tools/gradle-multi-project-decoupled-projects
# org.gradle.parallel=true
# AndroidX package structure to make it clearer which packages are bundled with the
# Android operating system, and which are packaged with your app's APK
# https://developer.android.com/topic/libraries/support-library/androidx-rn
android.useAndroidX=true
# Enables namespacing of each library's R class so that its R class includes only the
```

```
# resources declared in the library itself and none from the library's dependencies,
# thereby reducing the size of the R class for that library
android.nonTransitiveRClass=true
android.nonFinalResIds=false
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="Option Menu"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.DayNight.DarkActionBar"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

```
package com.example.optionmenu;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }  // END of onCrete()
    public boolean onCreateOptionsMenu(Menu menu)
    {
        getMenuInflater().inflate(R.menu.optionmenubar,menu);
        return true;
```

```
    }
    public boolean onOptionsItemSelected(MenuItem item)
    {
        int item_id=item.getItemId();
        switch (item_id)
        {
            case R.id.o1:
                Toast.makeText(this,"option 1 BCA",Toast.LENGTH_LONG).show();
                return  true;
            case R.id.o2:
                Toast.makeText(this,"option 2 BCS",Toast.LENGTH_LONG).show();
                return  true;
            case R.id.o3:
                Toast.makeText(this,"option 3 BCOM",Toast.LENGTH_LONG).show();
                return  true;
            default:
                return super.onOptionsItemSelected(item);
        }
    }




}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button"
        android:layout_width="214dp"
        android:layout_height="55dp"
        android:text="Department"
        android:textColor="#E20E0E"
        android:textColorLink="#E2D672"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.066"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.055" />

</androidx.constraintlayout.widget.ConstraintLayout>



popupmenu.xml

 <?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/o1"
        android:title="BCA" />
    <item
        android:id="@+id/o2"
```

```
        android:title="BCS" />
    <item
        android:id="@+id/o3"
        android:title="BCOM" />
</menu>
```

gradle.properties

```
 # Project-wide Gradle settings.
# IDE (e.g. Android Studio) users:
# Gradle settings configured through the IDE *will override*
# any settings specified in this file.
# For more details on how to configure your build environment visit
# http://www.gradle.org/docs/current/userguide/build_environment.html
# Specifies the JVM arguments used for the daemon process.
# The setting is particularly useful for tweaking memory settings.
org.gradle.jvmargs=-Xmx2048m -Dfile.encoding=UTF-8
# When configured, Gradle will run in incubating parallel mode.
# This option should only be used with decoupled projects. For more details, visit
# https://developer.android.com/r/tools/gradle-multi-project-decoupled-projects
# org.gradle.parallel=true
# AndroidX package structure to make it clearer which packages are bundled with the
# Android operating system, and which are packaged with your app's APK
# https://developer.android.com/topic/libraries/support-library/androidx-rn
android.useAndroidX=true
# Enables namespacing of each library's R class so that its R class includes only the
# resources declared in the library itself and none from the library's dependencies,
# thereby reducing the size of the R class for that library
android.nonTransitiveRClass=true
android.nonFinalResIds=false
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

```
package com.example.popupmenu;

import android.os.Bundle;
```

```
import android.view.MenuItem;
import android.view.View;
import android.widget.Button;
import android.widget.PopupMenu;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        btn=(Button)findViewById(R.id.button);
        btn.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v) {
                PopupMenu popupMenu=new PopupMenu(MainActivity.this,btn);
               // popupMenu.getMenuInflater().inflate(R.menu.popupmenu,popupMenu.getMenu());
getMenuInflater().inflate(R.menu.popupmenu,popupMenu.getMenu());
                popupMenu.setOnMenuItemClickListener(new PopupMenu.OnMenuItemClickListener()
                {
                    @Override
                    public boolean onMenuItemClick(MenuItem menuItem)
                    {
                        int item_id=menuItem.getItemId();
                        switch (item_id)
                        {
                            case R.id.o1:
                                Toast.makeText(MainActivity.this,"BCA
Department",Toast.LENGTH_LONG).show();
                                return  true;
                            case R.id.o2:
                                Toast.makeText(MainActivity.this,"BCS
Department",Toast.LENGTH_LONG).show();
                                return  true;
                            case R.id.o3:
                                Toast.makeText(MainActivity.this,"BCOM
Department",Toast.LENGTH_LONG).show();
                                return  true;
                            default:
                                return true;
                        }
                    }
                });

                popupMenu.show();
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
```

```
    }





}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="224dp"
        android:layout_height="91dp"
        android:text="Long Press"
        android:textColor="#B31111"
        android:textSize="34sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.582"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.254" />
</androidx.constraintlayout.widget.ConstraintLayout>


contextmenu.xml

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/o1"
        android:title="BCA" />
    <item
        android:id="@+id/o2"
        android:title="BCS" />
    <item
        android:id="@+id/o3"
        android:title="BCOM" />
</menu>


gradle.properties

 # Project-wide Gradle settings.
# IDE (e.g. Android Studio) users:
# Gradle settings configured through the IDE *will override*
# any settings specified in this file.
# For more details on how to configure your build environment visit
# http://www.gradle.org/docs/current/userguide/build_environment.html
# Specifies the JVM arguments used for the daemon process.
# The setting is particularly useful for tweaking memory settings.
```

```
org.gradle.jvmargs=-Xmx2048m -Dfile.encoding=UTF-8
# When configured, Gradle will run in incubating parallel mode.
# This option should only be used with decoupled projects. For more details, visit
# https://developer.android.com/r/tools/gradle-multi-project-decoupled-projects
# org.gradle.parallel=true
# AndroidX package structure to make it clearer which packages are bundled with the
# Android operating system, and which are packaged with your app's APK
# https://developer.android.com/topic/libraries/support-library/androidx-rn
android.useAndroidX=true
# Enables namespacing of each library's R class so that its R class includes only the
# resources declared in the library itself and none from the library's dependencies,
# thereby reducing the size of the R class for that library
android.nonTransitiveRClass=true
android.nonFinalResIds=false
```

AndroidManifest.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.ContextMenu"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

MainActivity.java

```java
package com.example.contextmenu;

import android.os.Bundle;
import android.view.ContextMenu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.view.View;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        TextView tv;
        tv=findViewById(R.id.textView);

        // Register a view for the context menu
        registerForContextMenu(tv);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
    public void onCreateContextMenu(ContextMenu menu, View v,
                                    ContextMenu.ContextMenuInfo menuInfo) {
       /* MenuInflater inflater = getMenuInflater();
        inflater.inflate(R.menu.contextmenu, menu);
        super.onCreateContextMenu(menu, v, menuInfo); */
        getMenuInflater().inflate(R.menu.contextmenu,menu);
    }
    public boolean onContextItemSelected(MenuItem menuItem)
    {
        int item_id=menuItem.getItemId();
        switch (item_id)
        {
            case R.id.o1:
                Toast.makeText(MainActivity.this,"BCA Department",Toast.LENGTH_LONG).show();
                return  true;
            case R.id.o2:
                Toast.makeText(MainActivity.this,"BCS Department",Toast.LENGTH_LONG).show();
                return  true;
            case R.id.o3:
                Toast.makeText(MainActivity.this,"BCOM Department",Toast.LENGTH_LONG).show();
                return  true;
            default:
                return true;
        }

    }
}
```

Expected Output:

```
Option Menu: Tap 3-dot menu icon → Shows options with Toast on selection.
Popup Menu: Click "Department" button → Popup shows BCA, BCS, BCOM.
Context Menu: Long press "Long Press Here" → Context menu appears with options.
```

OR

Q2 — OR (Alternative Question)

Demonstrate Array Adapter using List View to display list of Country. **[20 Marks]**

Program Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
    tools:context=".MainActivity">

    <ListView
        android:id="@+id/listview"
        android:layout_width="409dp"
        android:layout_height="729dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraint Top_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

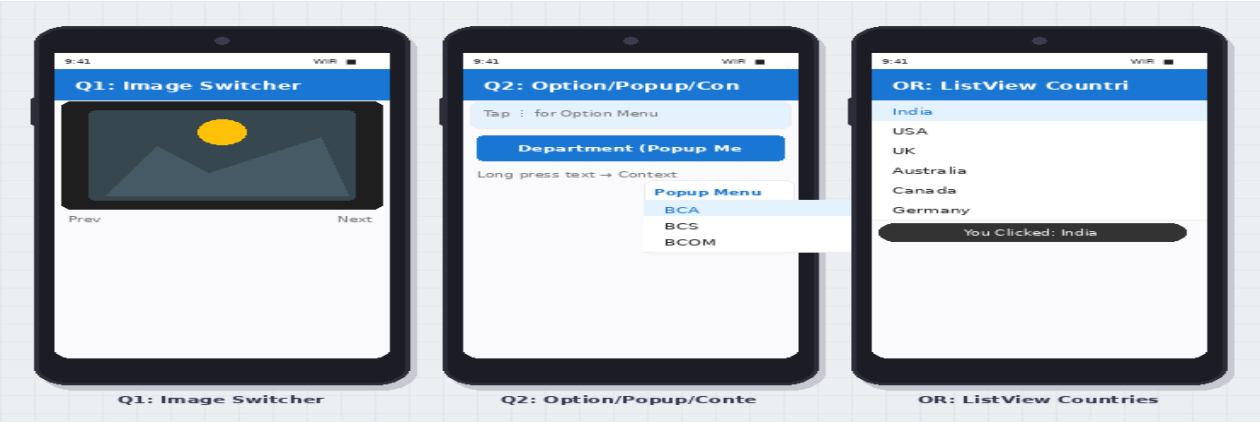
```
package com.example.listview;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    ListView l;
    String tutorials[] = { "India", "USA", "UK", "Australia", "Canada", "Germany", "Japan",
"France", "Brazil", "China" };

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        l = findViewById(R.id.listview); //support_simple_spinner_dropdown_item
        ArrayAdapter<String> arr= new ArrayAdapter<String>(this,
                androidx.appcompat.R.layout.support_simple_spinner_dropdown_item,tutorials);
        l.setAdapter(arr);

        l.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position,
                                    long id) {
                // Display a toast message with the clicked item text
                Toast.makeText(MainActivity.this, " You Clicked: " +
                        tutorials[position], Toast.LENGTH_SHORT).show();
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
```

```
    }
}
```

Expected Output:

```
ListView shows: India, USA, UK, Australia, Canada, Germany...
Click "India" → Toast: "You Clicked: India"
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)



Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 20

TY BCA (Science) Sem-VI | BCA 366: Android Programming | Duration: 3 Hrs | Max Marks: 50

Q1 — Question

Create an Android Application to accept two numbers and find power and Average. Display the result on the next activity on Button click. **[10 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="20dp" android:gravity="center">
    <EditText android:id="@+id/etBase" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Enter Base"
android:inputType="numberDecimal"/>
    <EditText android:id="@+id/etExp" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:hint="Enter Exponent"
android:inputType="numberDecimal"/>
    <Button android:id="@+id/btnCalculate" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Calculate"
android:layout_marginTop="16dp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.poweraverage;

import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText etBase = findViewById(R.id.etBase);
        EditText etExp  = findViewById(R.id.etExp);

        findViewById(R.id.btnCalculate).setOnClickListener(v -> {
            Intent i = new Intent(this, MainActivity2.class);
            i.putExtra("base", etBase.getText().toString());
            i.putExtra("exp",  etExp.getText().toString());
            startActivity(i);
        });
    }
}
```

activity_main2.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="24dp" android:gravity="center">
    <TextView android:id="@+id/tvPower" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:textSize="22sp"
        android:textStyle="bold" android:textColor="#1565C0"/>
    <TextView android:id="@+id/tvAverage" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:textSize="22sp"
        android:textStyle="bold" android:textColor="#2E7D32" android:layout_marginTop="16dp"/>
</LinearLayout>
```

MainActivity2.java

```java
package com.example.poweraverage;
```

```java
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        double base = Double.parseDouble(getIntent().getStringExtra("base"));
        double exp  = Double.parseDouble(getIntent().getStringExtra("exp"));
        double power   = Math.pow(base, exp);
        double average = (base + exp) / 2;
        ((TextView) findViewById(R.id.tvPower)).setText("Power   = " + power);
        ((TextView) findViewById(R.id.tvAverage)).setText("Average = " + average);
    }
}
```

Expected Output:

```
Enter Base: 2, Exponent: 3 → Click Calculate.
Activity 2 shows:
  Power   = 8.0
  Average = 2.5
```

Q2 — Question

Create an Android application that creates a custom Alert Dialog containing Friends Name and onClick of Friend Name Button greet accordingly. **[20 Marks]**

Program Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:gravity="center" android:orientation="vertical" android:padding="20dp">
    <Button android:id="@+id/btnShowFriends"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Show Friends List" android:textSize="18sp"/>
    <TextView android:id="@+id/tvGreeting"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:textSize="20sp" android:textStyle="bold" android:textColor="#4A148C"
        android:layout_marginTop="24dp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.example.friendsdialog;

import android.content.DialogInterface;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    final String[] friends = {"Alice", "Bob", "Charlie", "Diana"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        TextView tvGreeting = findViewById(R.id.tvGreeting);
```

```
        findViewById(R.id.btnShowFriends).setOnClickListener(v -> {
            AlertDialog.Builder builder = new AlertDialog.Builder(this);
            builder.setTitle("Select a Friend");
            builder.setItems(friends, (dialog, which) ->
                tvGreeting.setText("Hello, " + friends[which] + "! How are you?"));
            builder.setNegativeButton("Cancel", null);
            builder.show();
        });
    }
}
```

Expected Output:

```
Click "Show Friends List" → Custom AlertDialog with items: Alice, Bob, Charlie, Diana.
Click "Bob" → Dialog closes → TextView: "Hello, Bob! How are you?"
```

OR

Q2 — OR (Alternative Question)

Create an Android Application to perform Zoom In, Zoom Out operation and display Satellite view, on Google Map. **[20 Marks]**

Program Code:

MapsActivity.java

```
package com.example.googlemapzoom;

import androidx.fragment.app.FragmentActivity;
import android.os.Bundle;
import android.widget.Button;
import com.google.android.gms.maps.*;
import com.google.android.gms.maps.model.*;

public class MapsActivity extends FragmentActivity implements OnMapReadyCallback {
    private GoogleMap mMap;
    private float zoomLevel = 12f;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_maps);
        SupportMapFragment mapFragment = (SupportMapFragment)
            getSupportFragmentManager().findFragmentById(R.id.map);
        mapFragment.getMapAsync(this);

        ((Button) findViewById(R.id.btnZoomIn)).setOnClickListener(v -> {
            zoomLevel = Math.min(zoomLevel + 1, 21);
            if (mMap != null) mMap.animateCamera(CameraUpdateFactory.zoomTo(zoomLevel));
        });
        ((Button) findViewById(R.id.btnZoomOut)).setOnClickListener(v -> {
            zoomLevel = Math.max(zoomLevel - 1, 1);
            if (mMap != null) mMap.animateCamera(CameraUpdateFactory.zoomTo(zoomLevel));
        });
    }

    @Override
    public void onMapReady(GoogleMap googleMap) {
        mMap = googleMap;
        mMap.setMapType(GoogleMap.MAP_TYPE_SATELLITE); // Satellite view
        LatLng pune = new LatLng(18.5204, 73.8567);
        mMap.addMarker(new MarkerOptions().position(pune).title("Pune"));
        mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(pune, zoomLevel));
    }
}
```

Expected Output:

```
Map loads in Satellite mode centred on Pune.
Click "Zoom In +" → Map zooms in (zoom level increases by 1).
Click "Zoom Out -" → Map zooms out (zoom level decreases by 1).
```

Expected Output Screens (Q1 left · Q2 center · Q2 OR right)

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]